STT 0626 - Pesquisa Operacional para Sistemas Logísticos e Engenharia de Transportes

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Aula anterior

Madeira



Minério de ferro



Aula anterior

x	y	Z
0	0	0
0	6	36
3,2	4,8	41,6
6	2	36
7	0	28

3.2 toneladas de minério

4.8 toneladas de madeira

Lucro de 41.6 UM

Aula anterior

x	y	Z
0	0	0
0	6	36
3,2	4,8	41,6
6	2	36
7	0	28

Valores fracionários

3.2 toneladas de minério

4.8 toneladas de madeira

Lucro de 41.6 UM

Programação linear inteira



Programação linear inteira



X1



Programação linear inteira



X1



X2



Programação linear inteira



X1



X2



$$Z = 6.3\ x1 + 7.8\ x2$$

Programação linear inteira



X1



X2



~~Z = 6.3 x1 + 7.8 x2~~

Não comporta valores fracionários

Programação linear inteira

- Utilização de equipamentos



X = 3.44 navios cargueiros
- não tem significado prático

Programação linear inteira

- Utilização de equipamentos

Programação linear inteira

- Utilização de equipamentos
- Tamanhos de lotes

Programação linear inteira

- Utilização de equipamentos
- Tamanhos de lotes



X = 1.35 kg

Programação linear inteira

- Utilização de equipamentos
- Tamanhos de lotes



X = 1.35 kg



X = 5 pacotes

$Z = 5x1 + 8x2$

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$



$Z = 5x1 + 8x2$

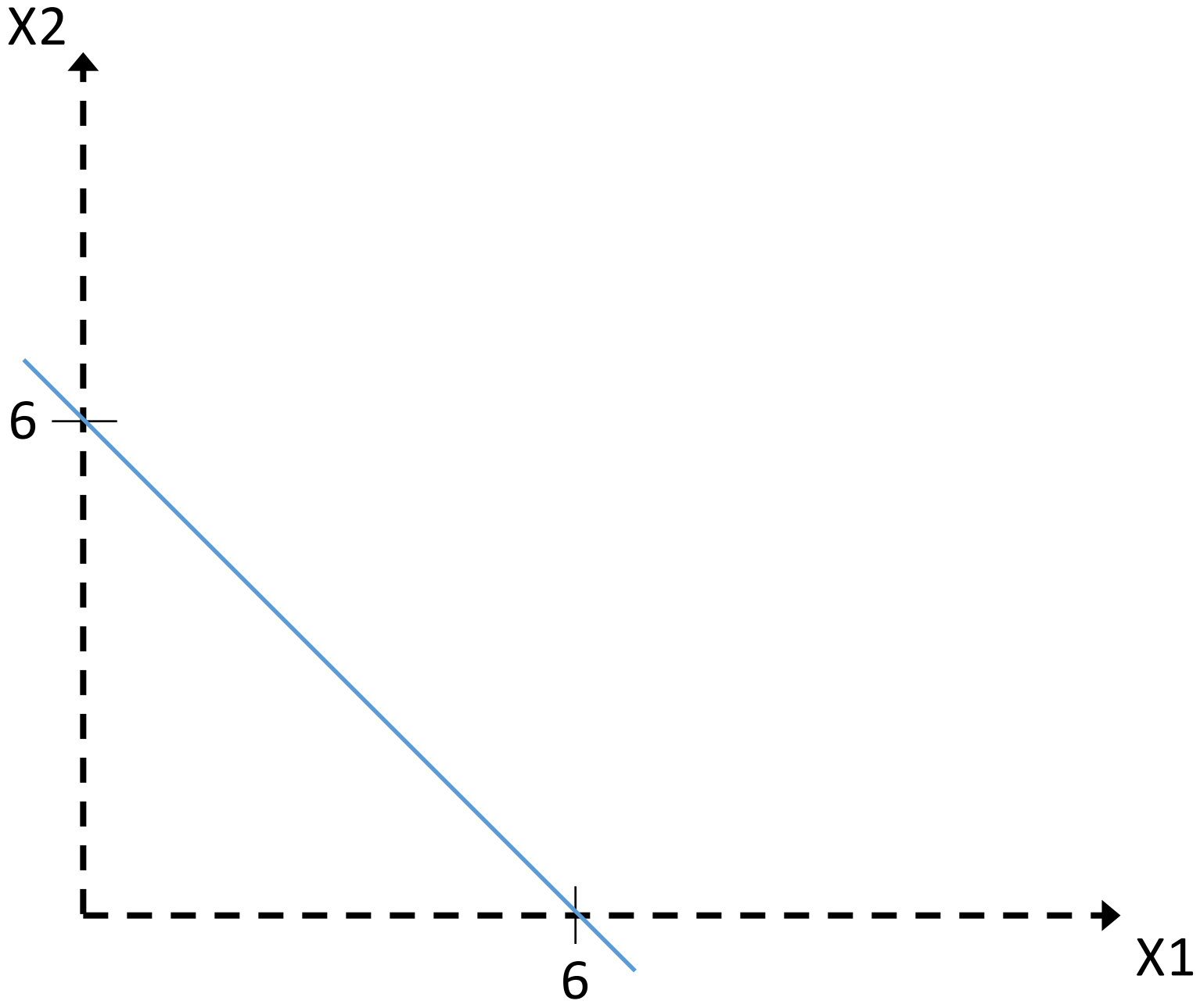
$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

$X1 + x2 = 6$

$X1=0;\ x2=6$

$X2=0;\ x1=6$



$Z = 5x1 + 8x2$

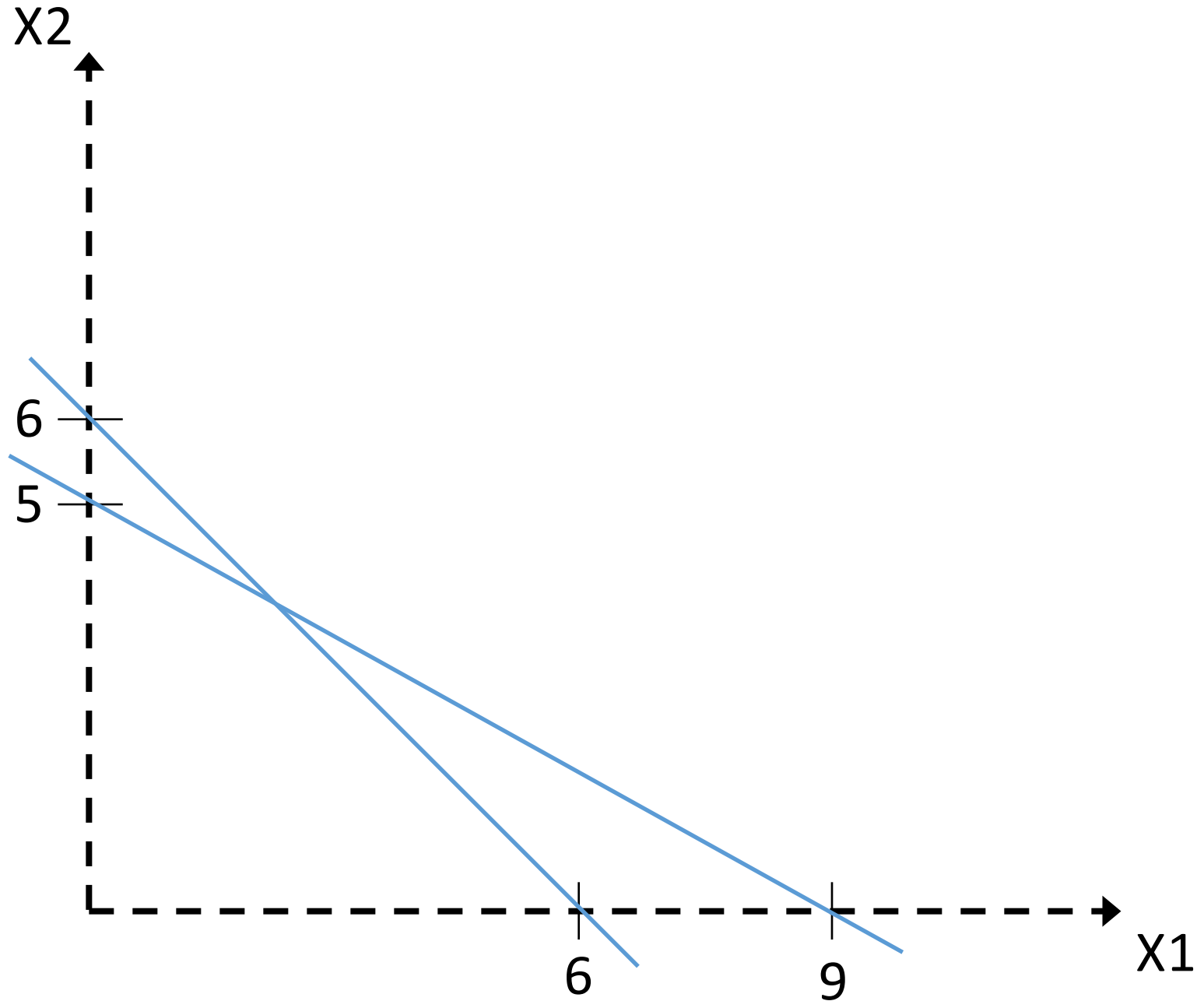
$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

$5x1 + 9x2 = 45$

$X1=0;\ x2=5$

$X2=0;\ x1=9$



$Z = 5x1 + 8x2$

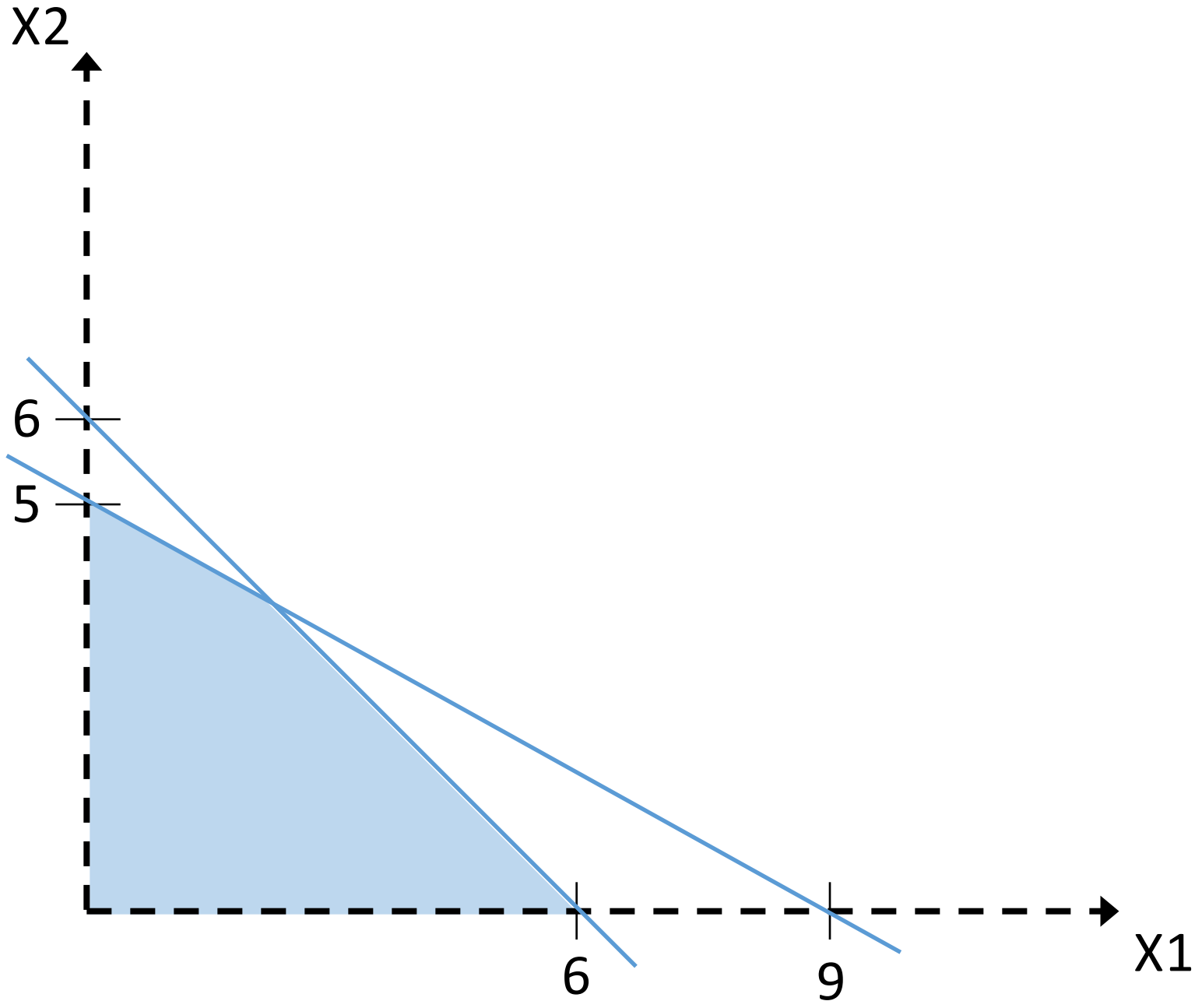
$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

$5x1 + 9x2 = 45$

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$X2=0;\ x1=9$



$Z = 5x1 + 8x2$

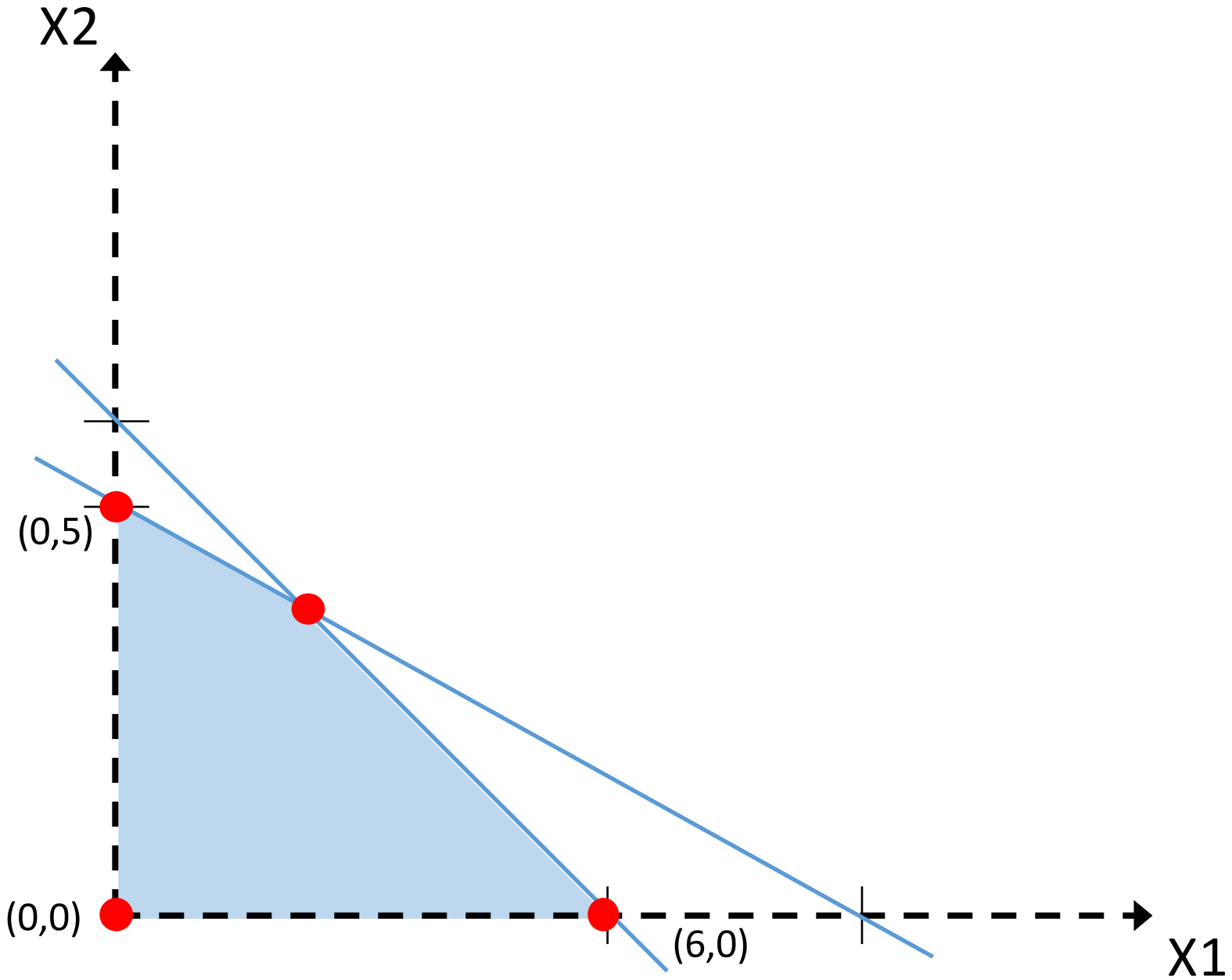
$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

$5x1 + 9x2 = 45$

$X1=0;\ x2=5$

$X2=0;\ x1=9$



$Z = 5x1 + 8x2$

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

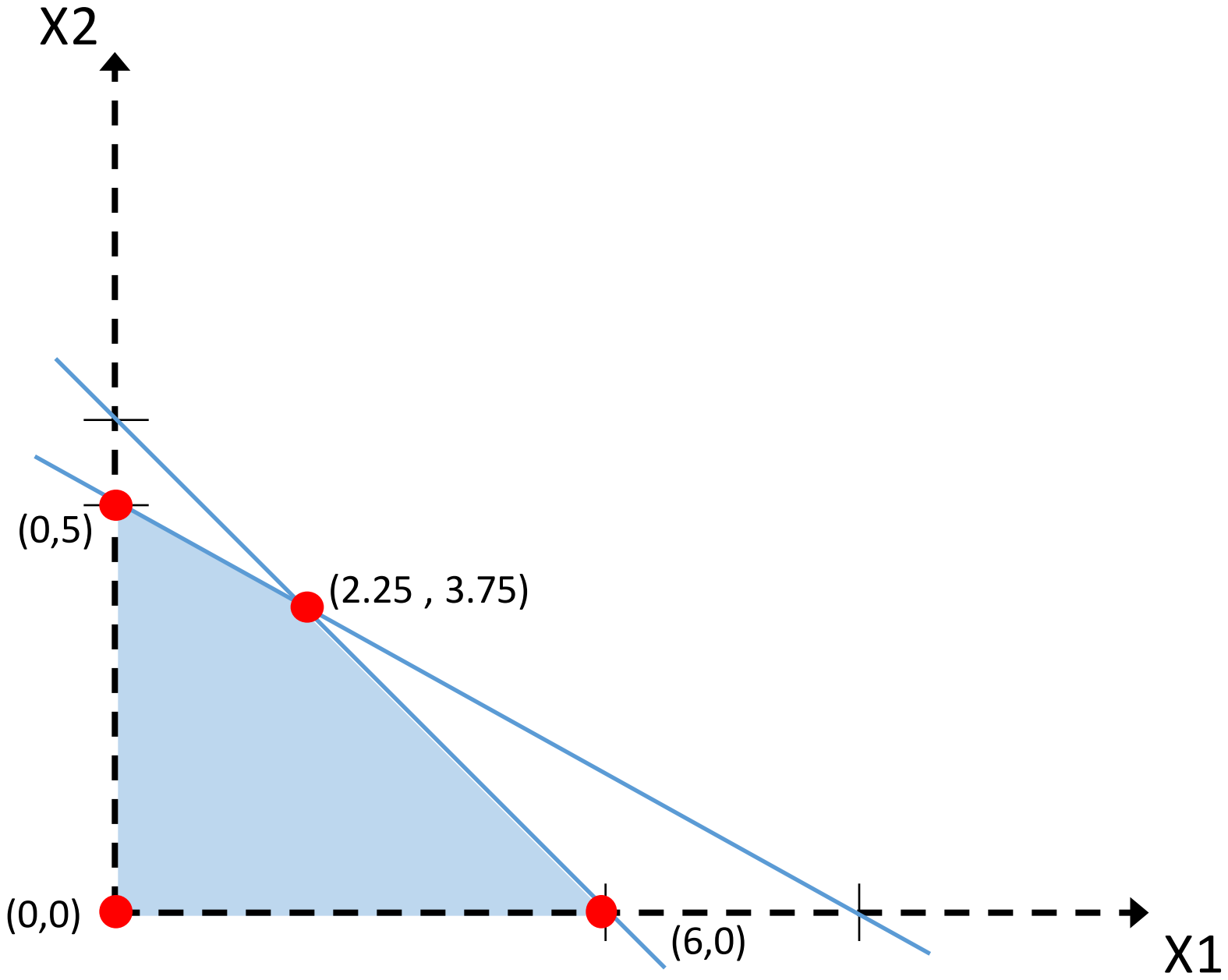
$X1 = 6 - x2$

$30 - 5x2 + 9x2 = 45$

$4x2 = 15$

$\underline{X2 = 3.75}$

$\underline{X1 = 2.25}$

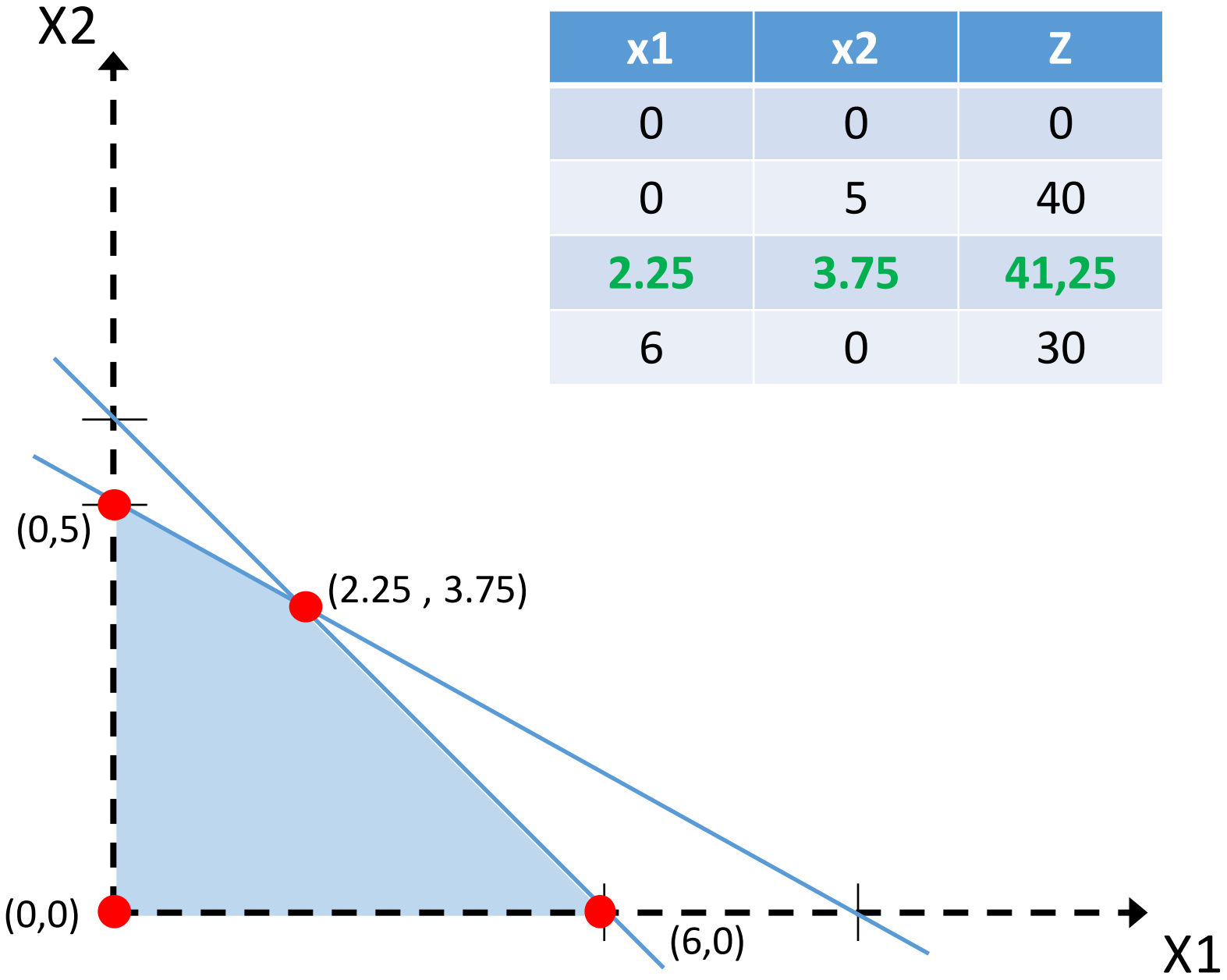


$Z = 5x1 + 8x2$

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41,25**
6	0	30



$Z = 5x1 + 8x2$

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

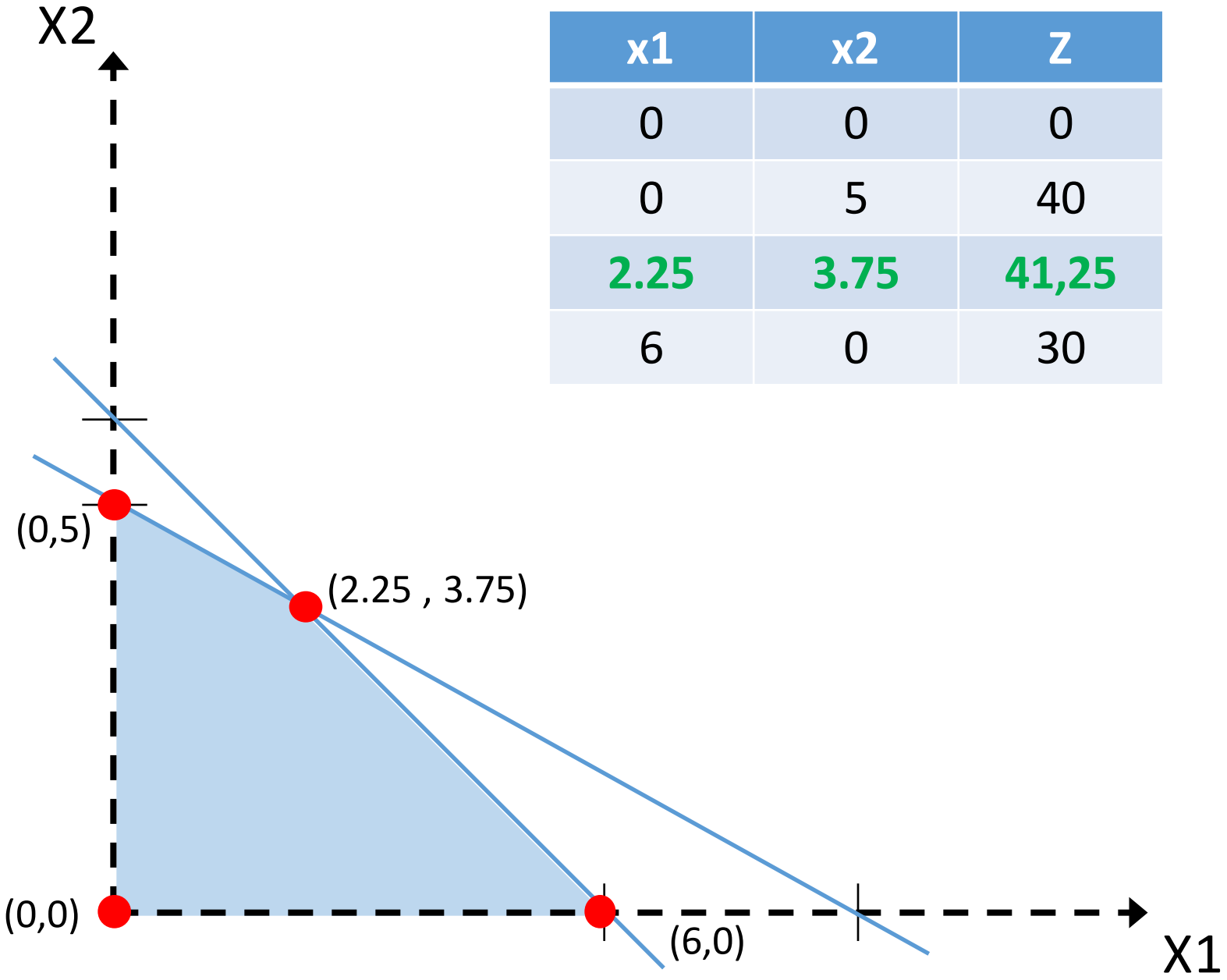
Truncamento

$x1 = 2$

$x2 = 3$

$Z = 34$

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41,25**
6	0	30



$Z = 5x1 + 8x2$

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
6	0	30

Truncamento

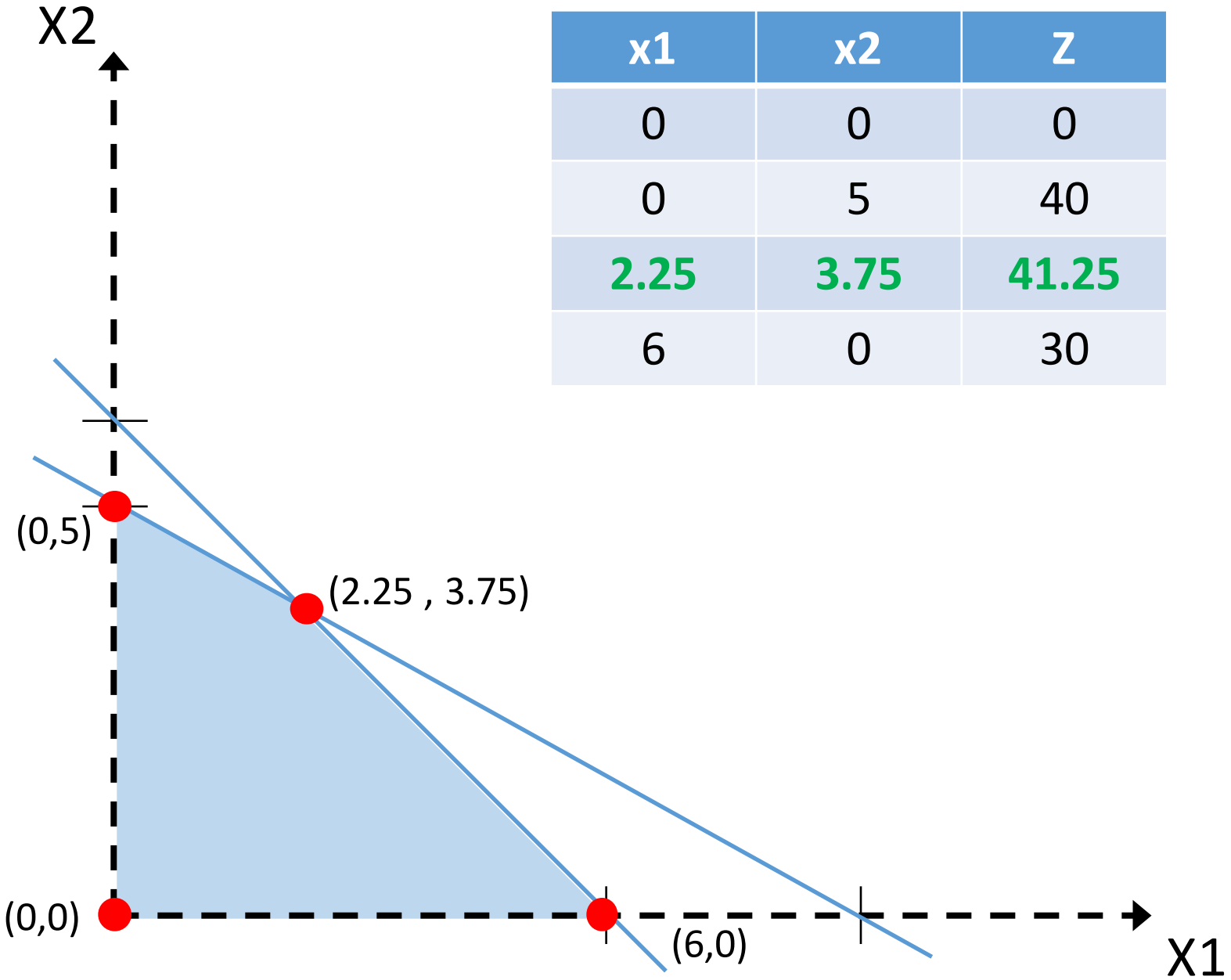
x1 = 2

x2 = 3

Z = 34

Tolerância

1-(34/41.25) ~ 0.18



$Z = 5x1 + 8x2$

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

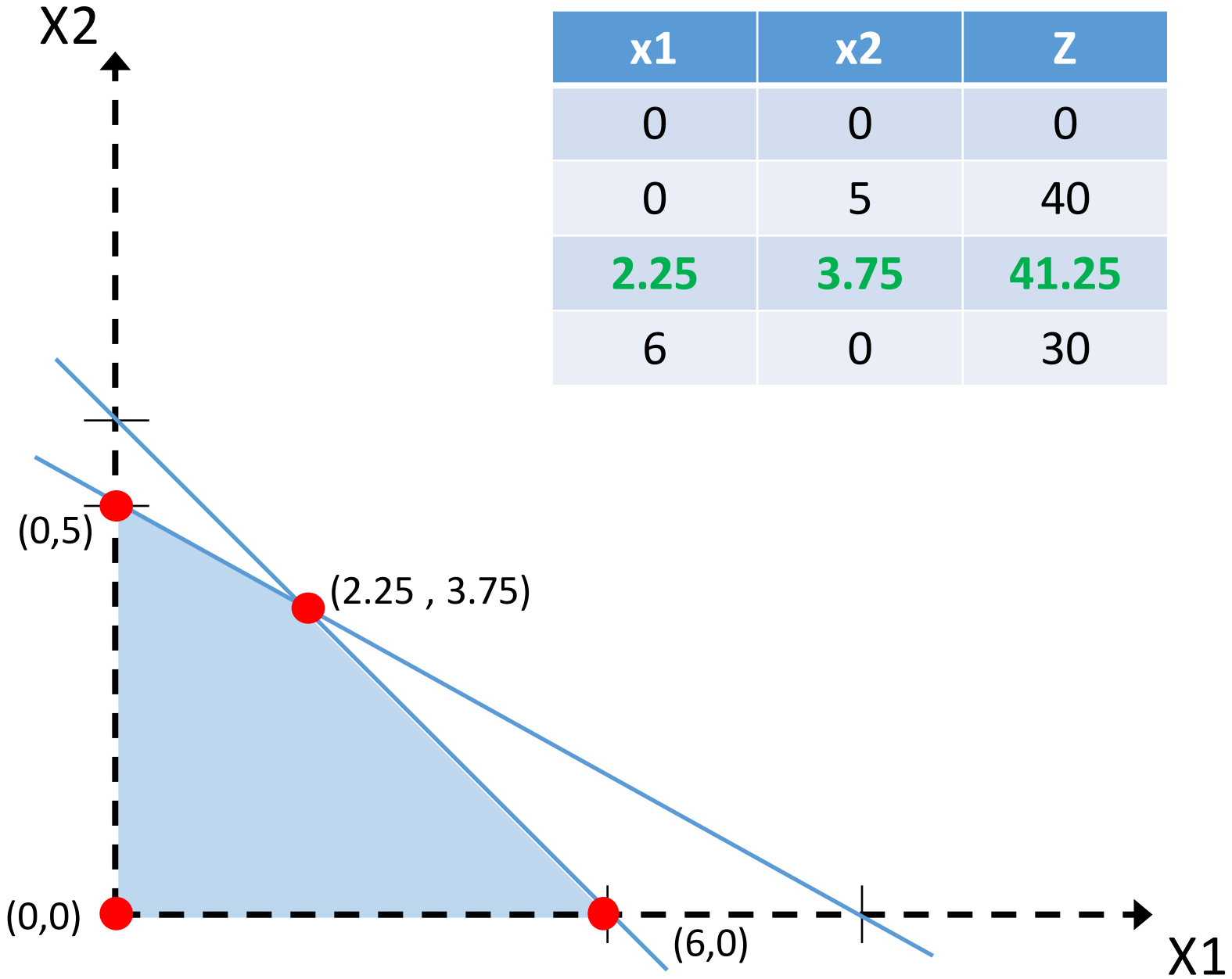
Truncamento

$x1 = 2$

$x2 = 4$

$Z = 42$!!!

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
6	0	30



$Z = 5x1 + 8x2$

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

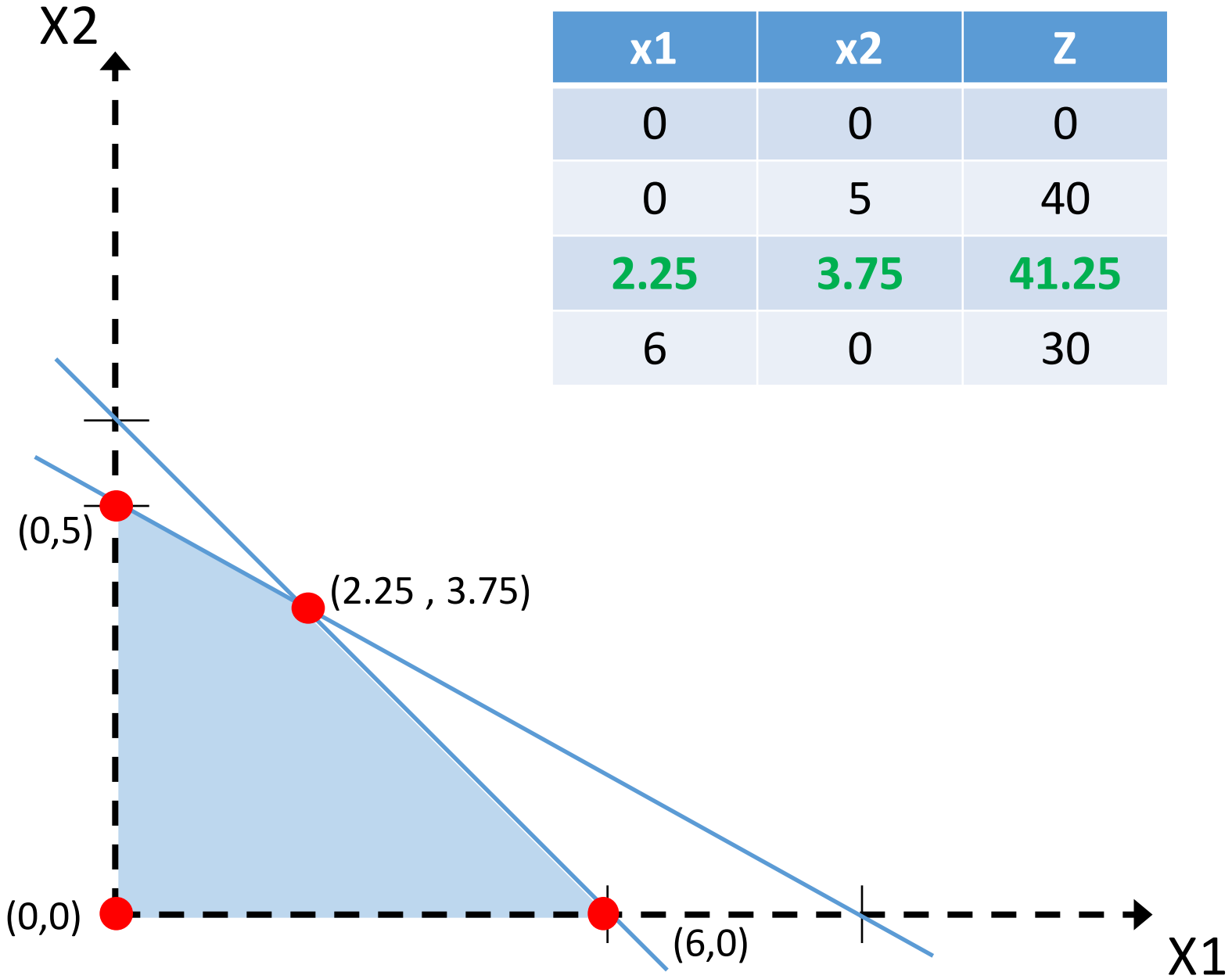
Truncamento

~~$x1 = 2$~~

~~$x2 = 4$~~

~~$Z = 42$!!!~~

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
6	0	30



$Z = 5x1 + 8x2$

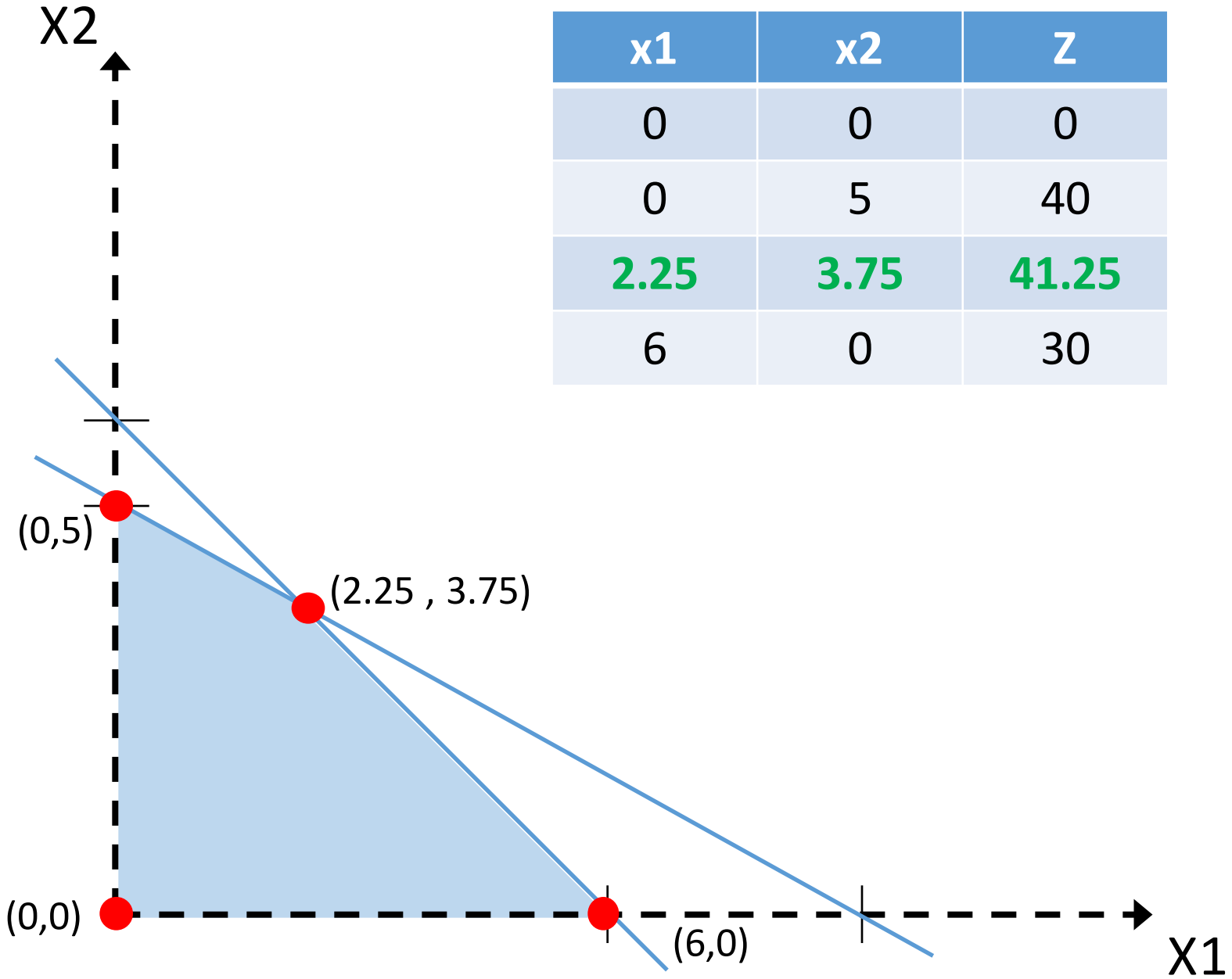
$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
6	0	30

Solução:

vars inteiras > vars relaxadas ?



$Z = 5x1 + 8x2$

$X1 + x2 \leq 6$

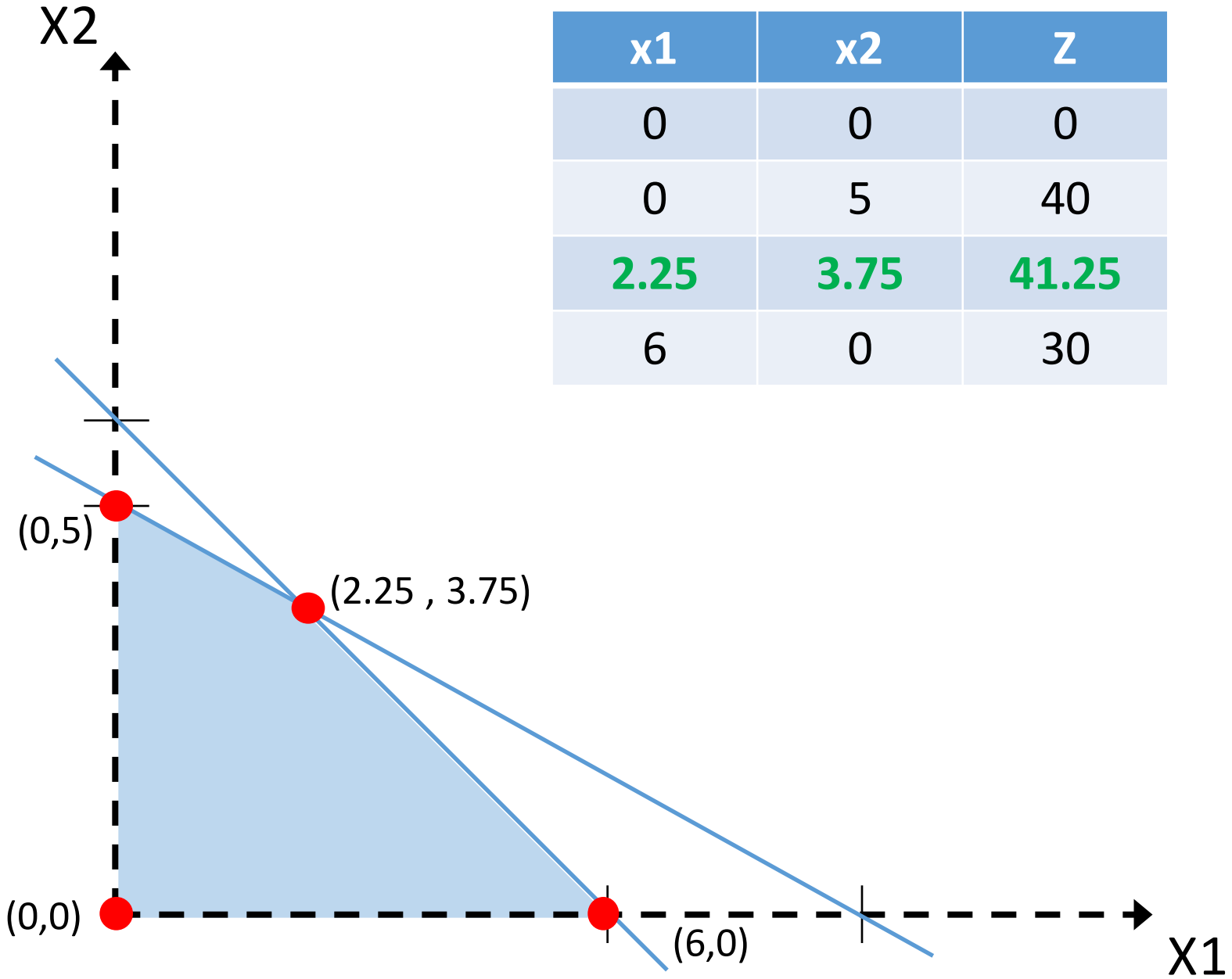
$5x1 + 9x2 \leq 45$

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
6	0	30

Solução:

vars inteiras > vars relaxadas ?

Não

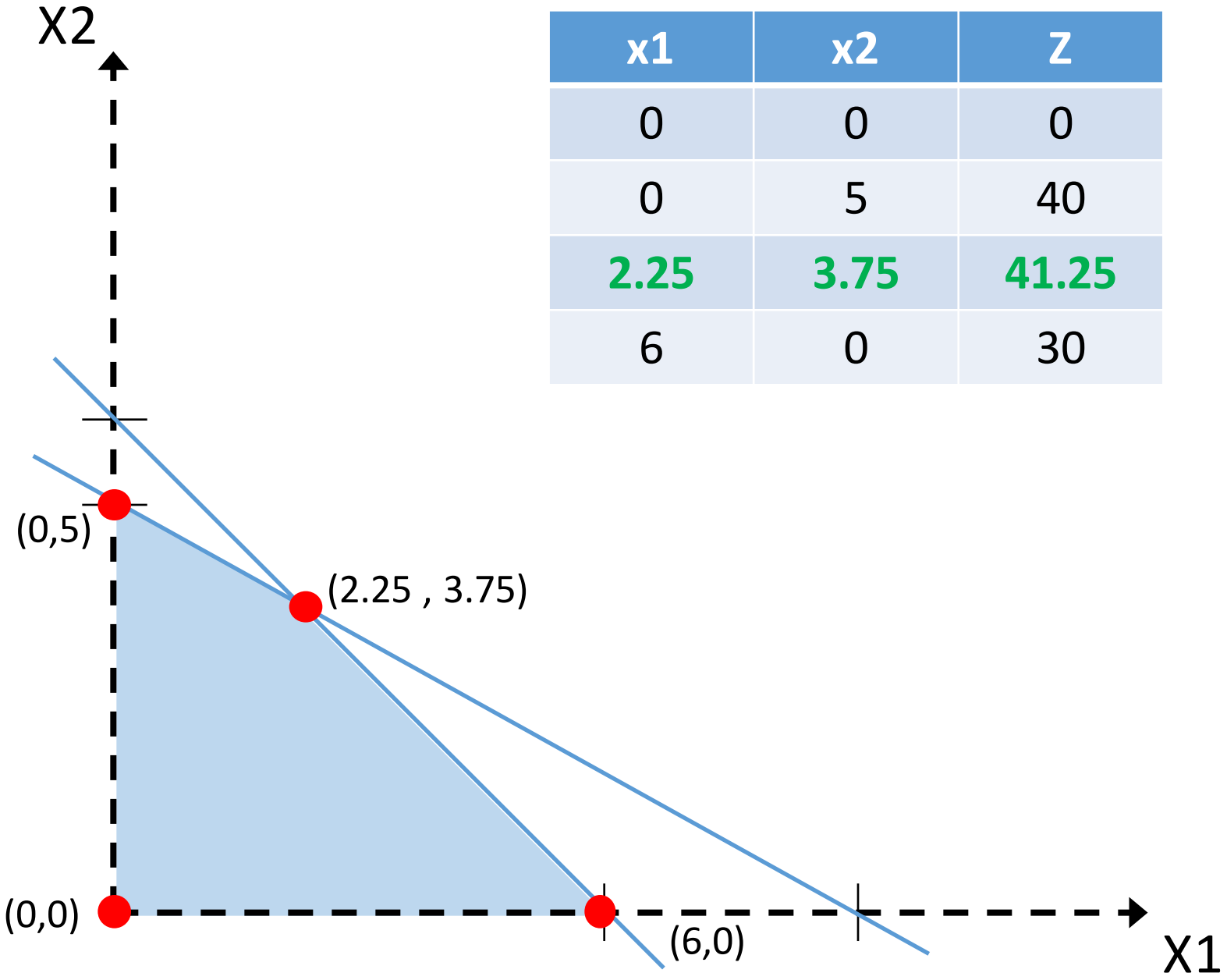


$Z = 5x1 + 8x2$

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
6	0	30



Solução:
vars inteiras > vars relaxadas ?

Não

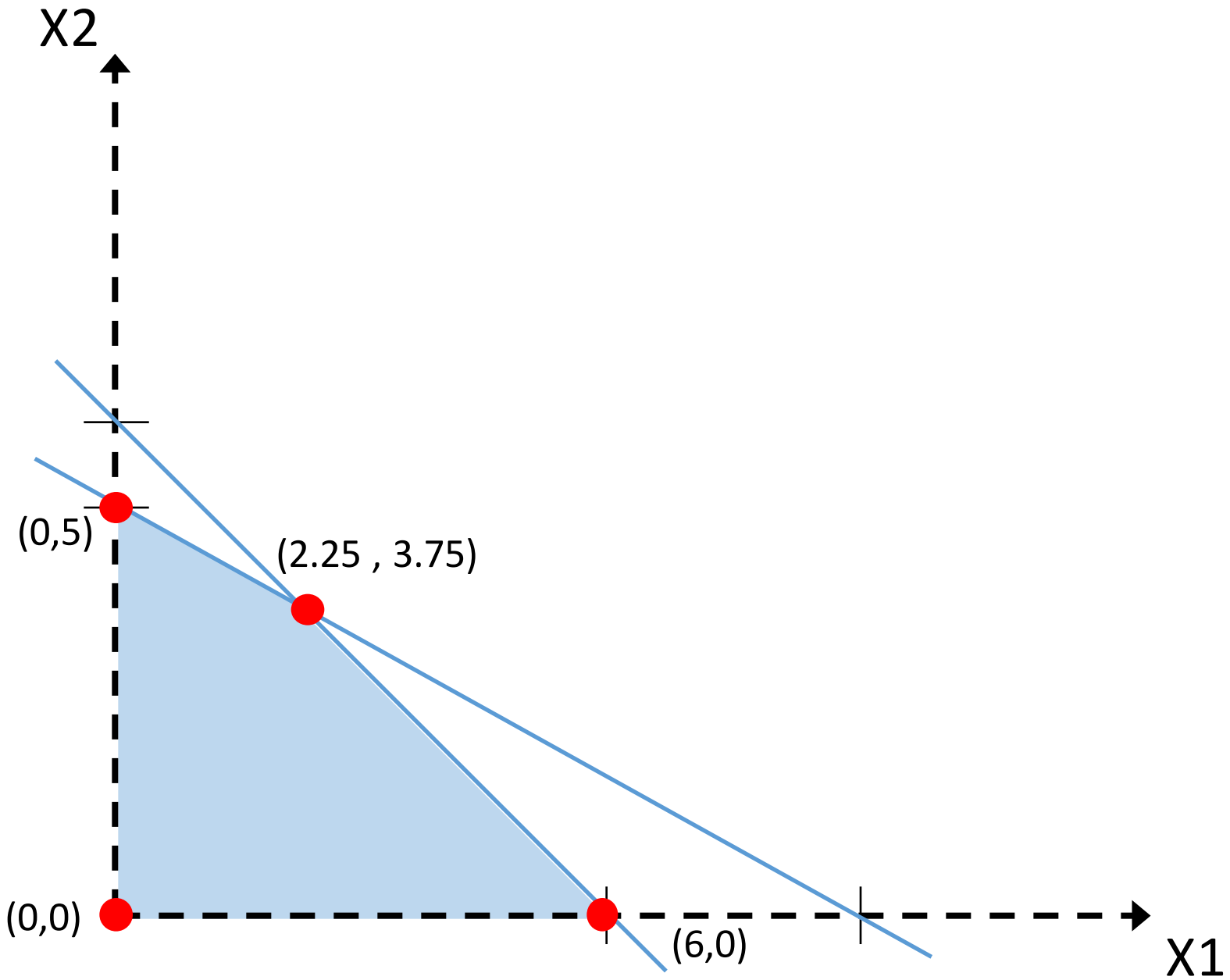
Vars relaxadas exploram o limite das restrições técnicas.

$Z = 5x1 + 8x2$ **41.25**

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
6	0	30

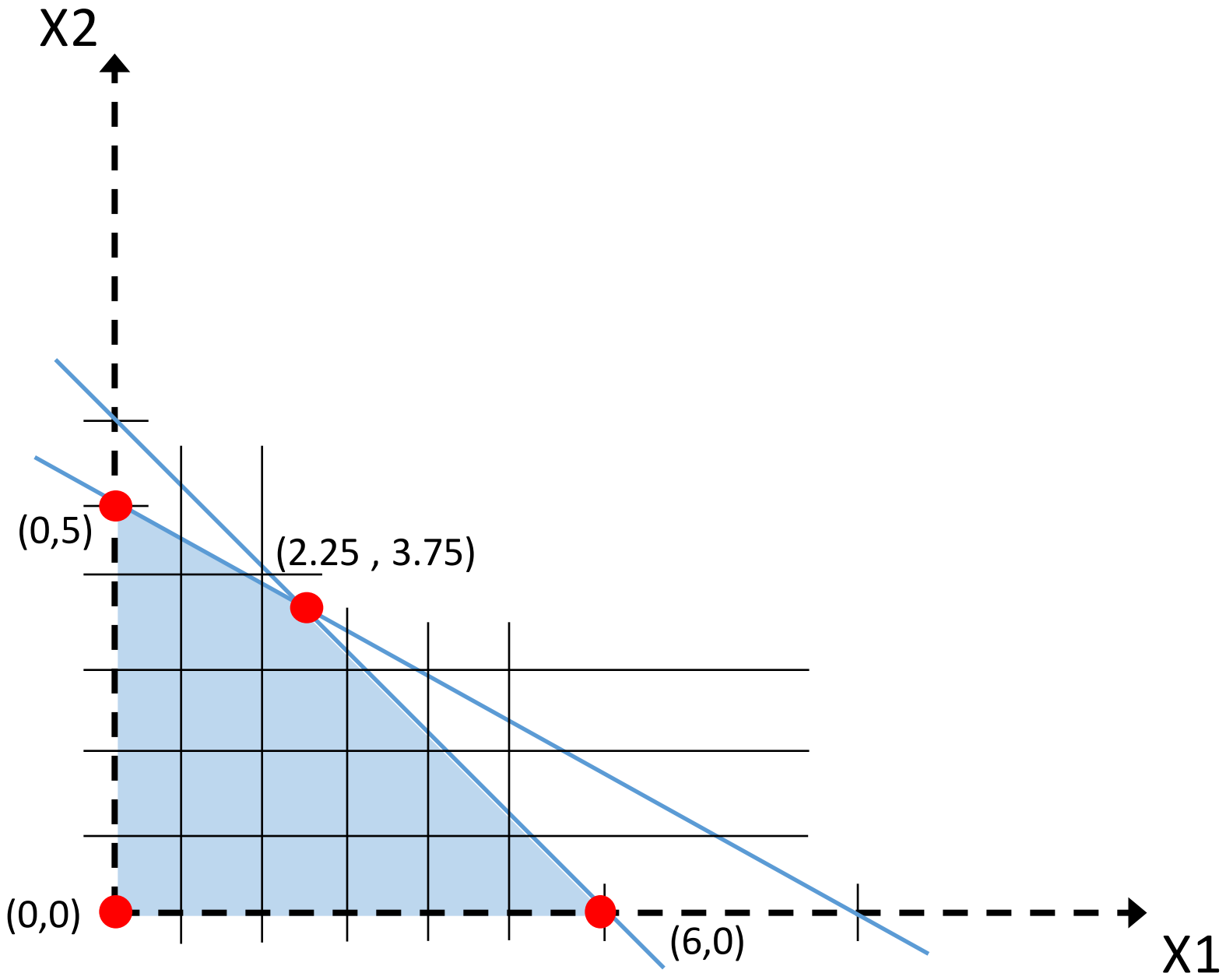


$Z = 5x1 + 8x2$ **41.25**

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
6	0	30

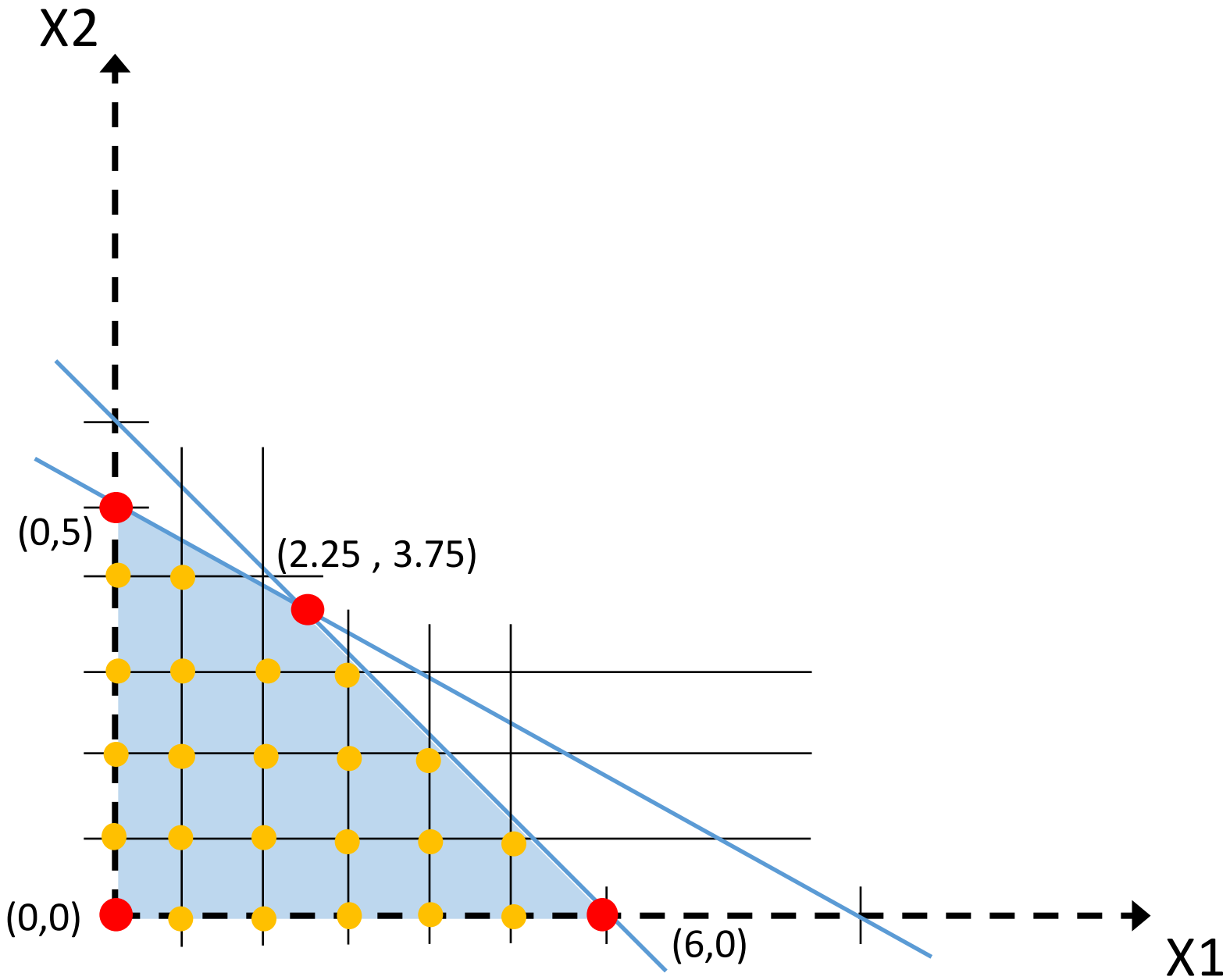


$Z = 5x1 + 8x2$ **41.25**

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
6	0	30



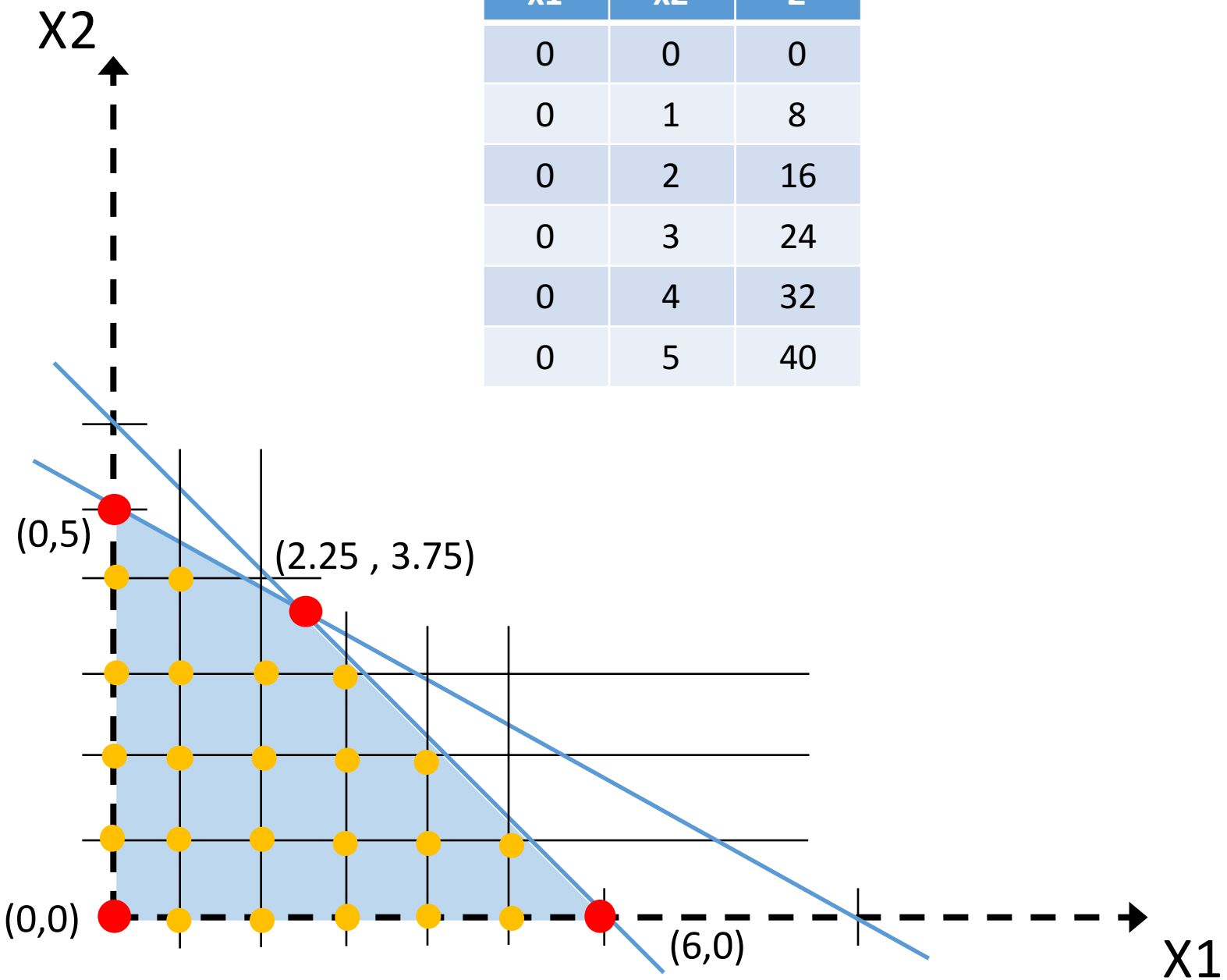
$Z = 5x1 + 8x2$ **41.25**

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
6	0	30

x1	x2	Z
0	0	0
0	1	8
0	2	16
0	3	24
0	4	32
0	5	40

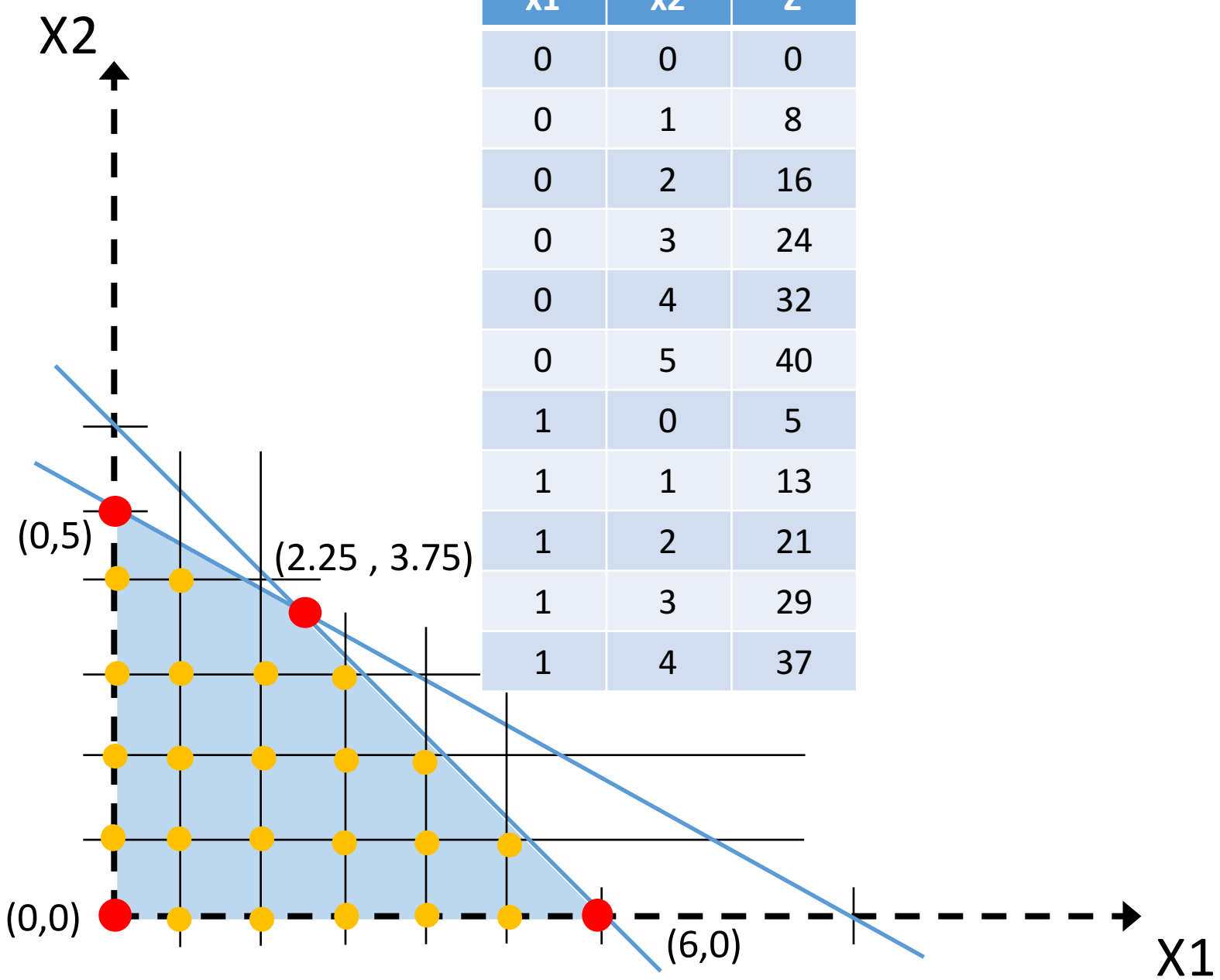


$Z = 5x1 + 8x2$ **41.25**

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
6	0	30



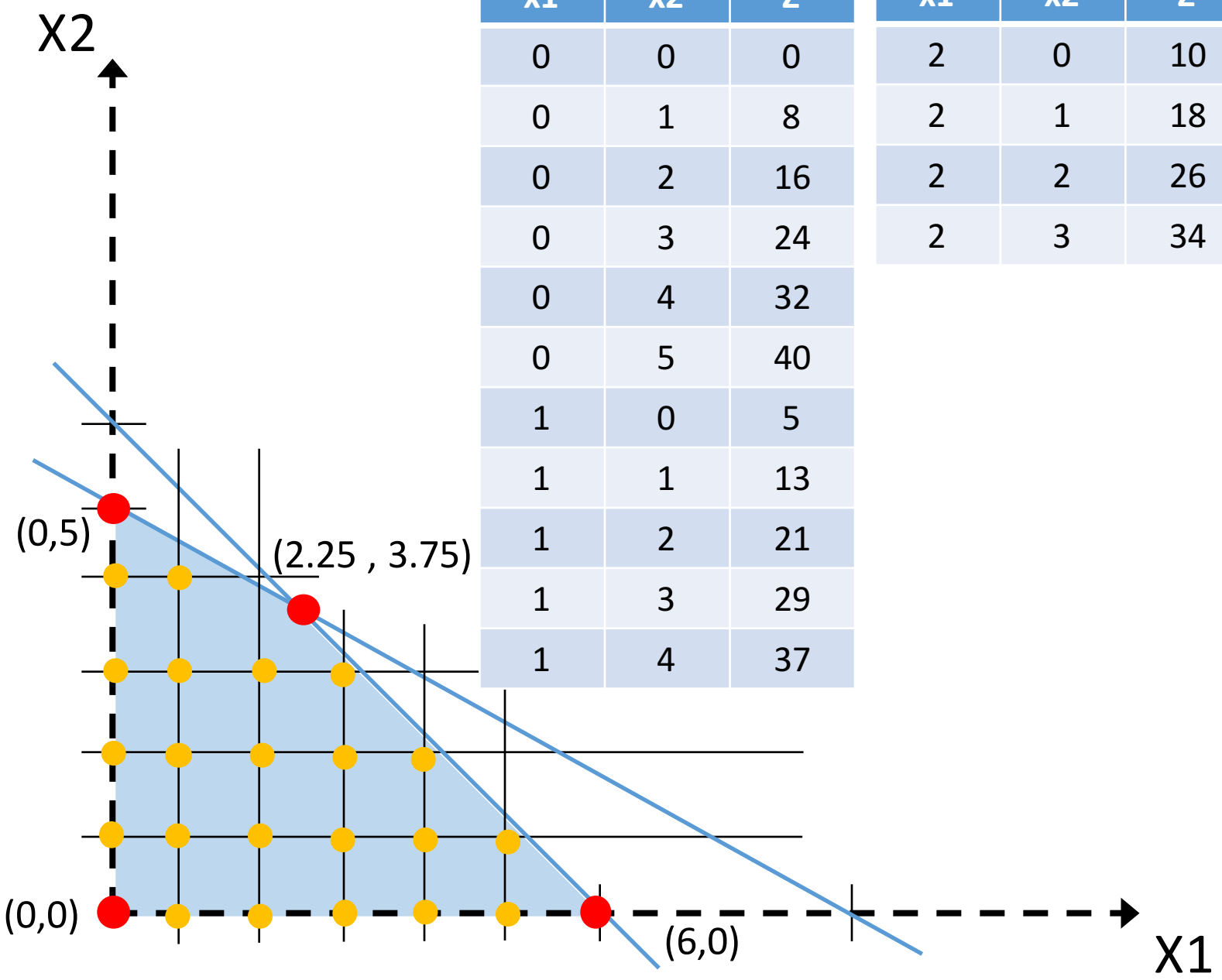
x1	x2	Z
0	0	0
0	1	8
0	2	16
0	3	24
0	4	32
0	5	40
1	0	5
1	1	13
1	2	21
1	3	29
1	4	37

$Z = 5x1 + 8x2$ **41.25**

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
6	0	30



x1	x2	Z
0	0	0
0	1	8
0	2	16
0	3	24
0	4	32
0	5	40
1	0	5
1	1	13
1	2	21
1	3	29
1	4	37

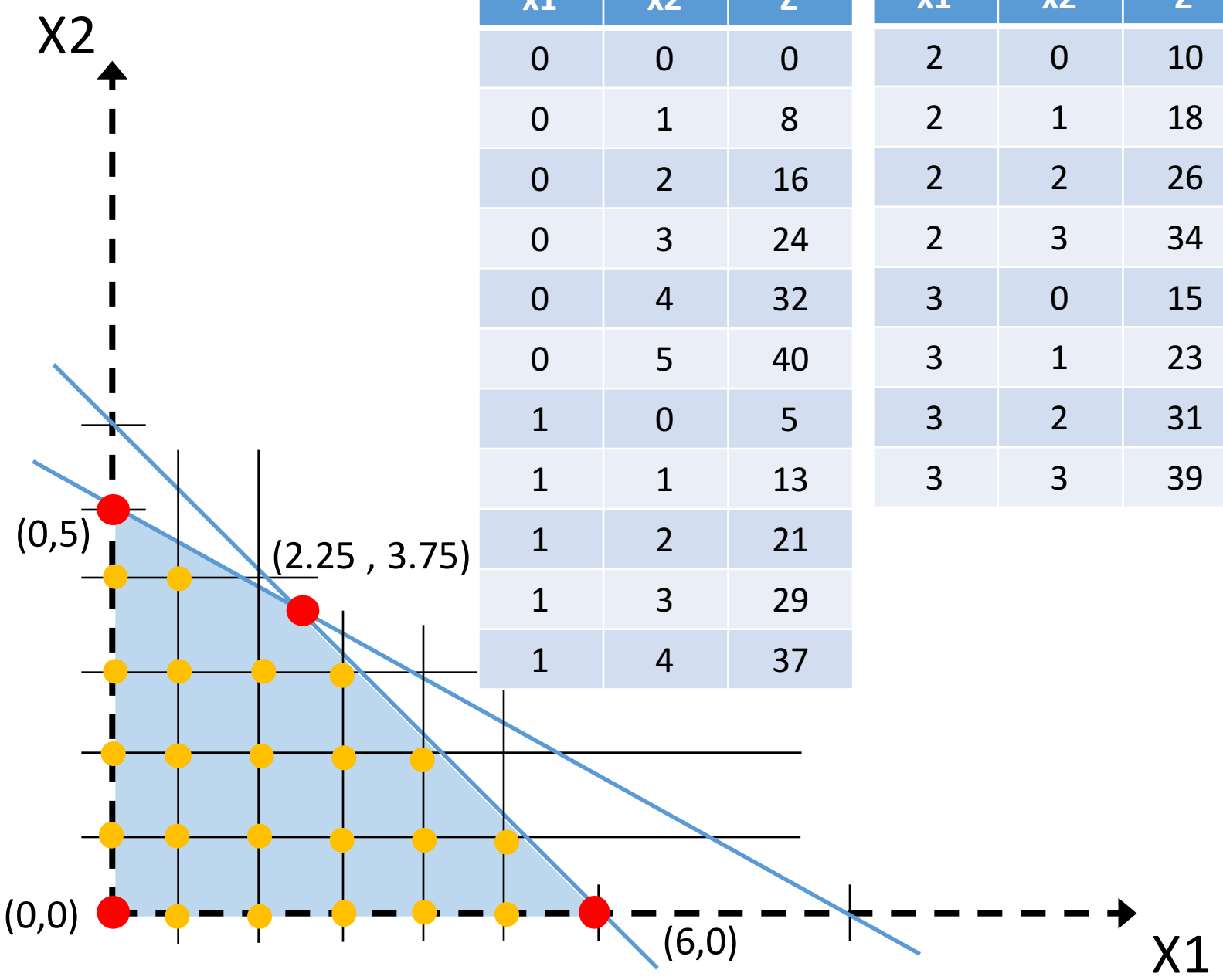
x1	x2	Z
2	0	10
2	1	18
2	2	26
2	3	34

$Z = 5x1 + 8x2$ **41.25**

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
6	0	30



x1	x2	Z
0	0	0
0	1	8
0	2	16
0	3	24
0	4	32
0	5	40
1	0	5
1	1	13
1	2	21
1	3	29
1	4	37

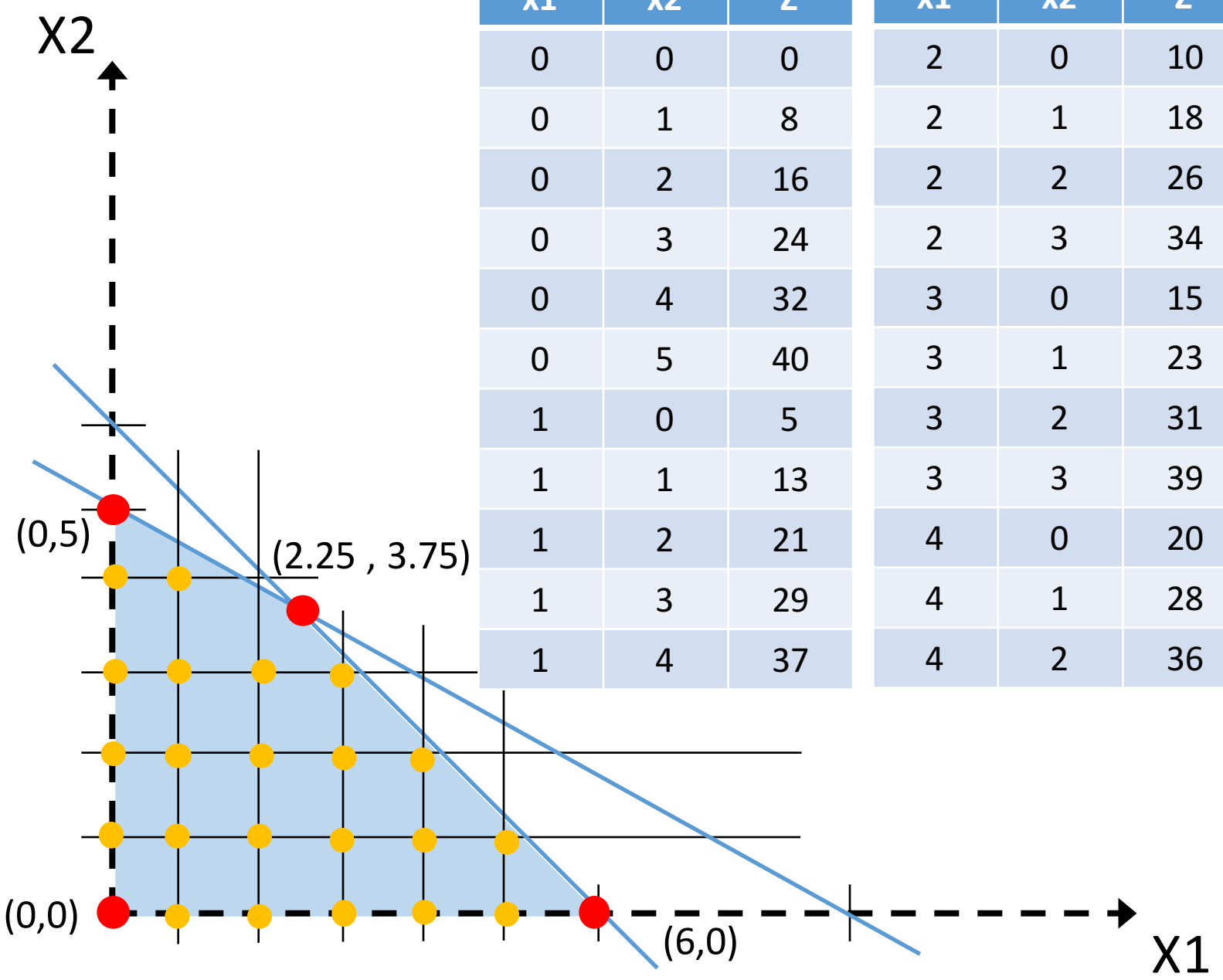
x1	x2	Z
2	0	10
2	1	18
2	2	26
2	3	34
3	0	15
3	1	23
3	2	31
3	3	39

$Z = 5x1 + 8x2$ **41.25**

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
6	0	30



x1	x2	Z
0	0	0
0	1	8
0	2	16
0	3	24
0	4	32
0	5	40
1	0	5
1	1	13
1	2	21
1	3	29
1	4	37

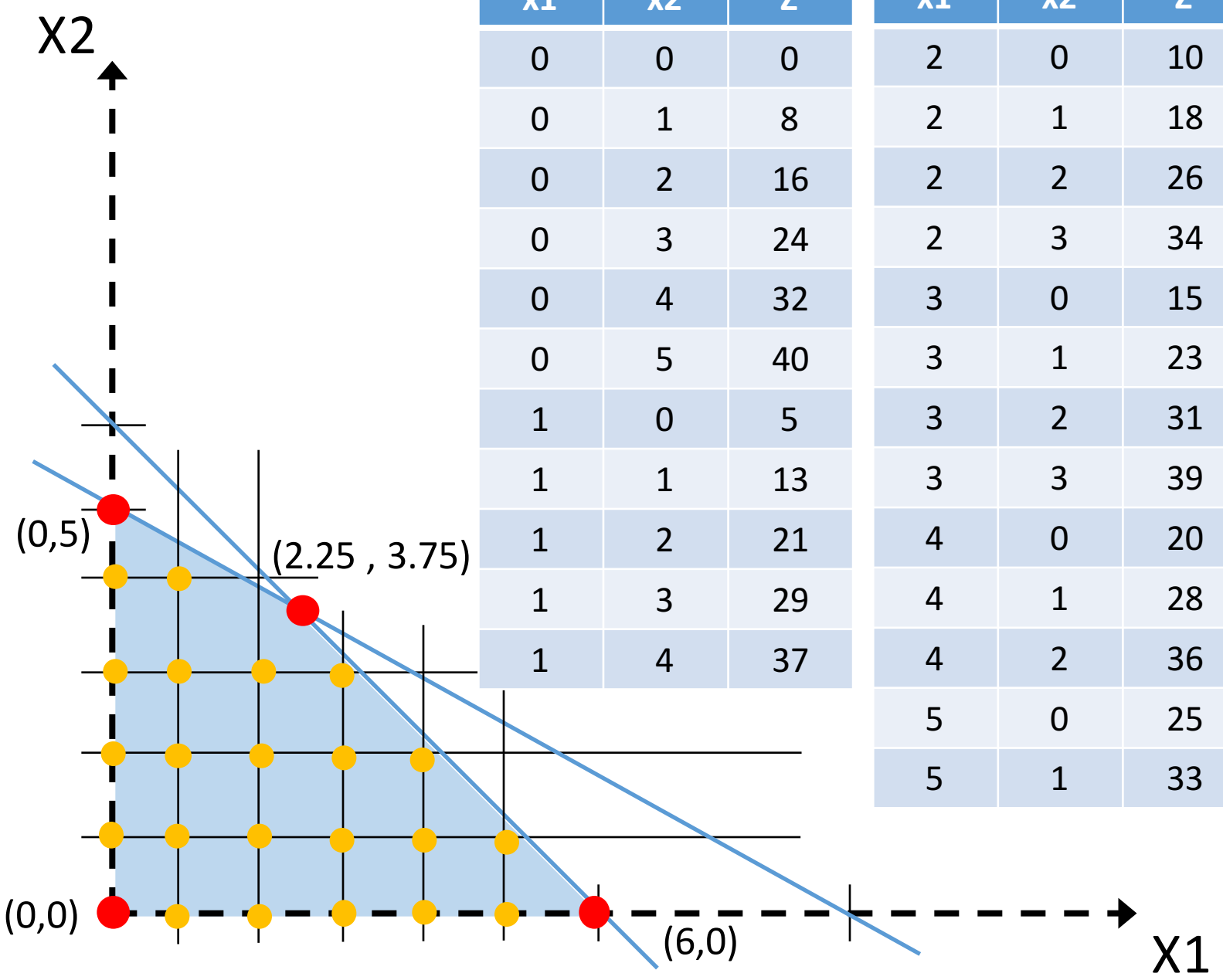
x1	x2	Z
2	0	10
2	1	18
2	2	26
2	3	34
3	0	15
3	1	23
3	2	31
3	3	39
4	0	20
4	1	28
4	2	36

$Z = 5x1 + 8x2$ **41.25**

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
6	0	30



x1	x2	Z
0	0	0
0	1	8
0	2	16
0	3	24
0	4	32
0	5	40
1	0	5
1	1	13
1	2	21
1	3	29
1	4	37

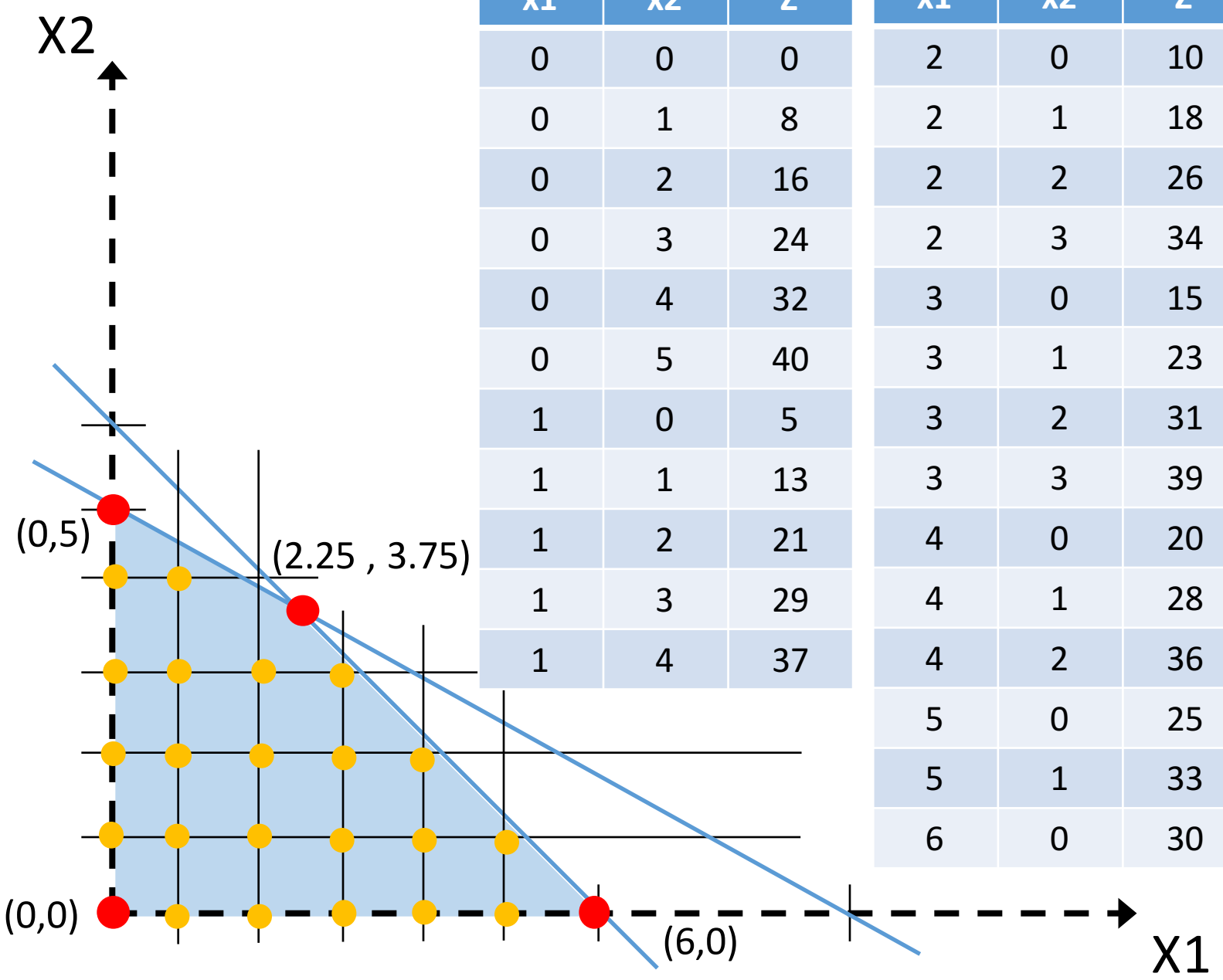
x1	x2	Z
2	0	10
2	1	18
2	2	26
2	3	34
3	0	15
3	1	23
3	2	31
3	3	39
4	0	20
4	1	28
4	2	36
5	0	25
5	1	33

$Z = 5x1 + 8x2$ **41.25**

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
6	0	30



x1	x2	Z
0	0	0
0	1	8
0	2	16
0	3	24
0	4	32
0	5	40
1	0	5
1	1	13
1	2	21
1	3	29
1	4	37

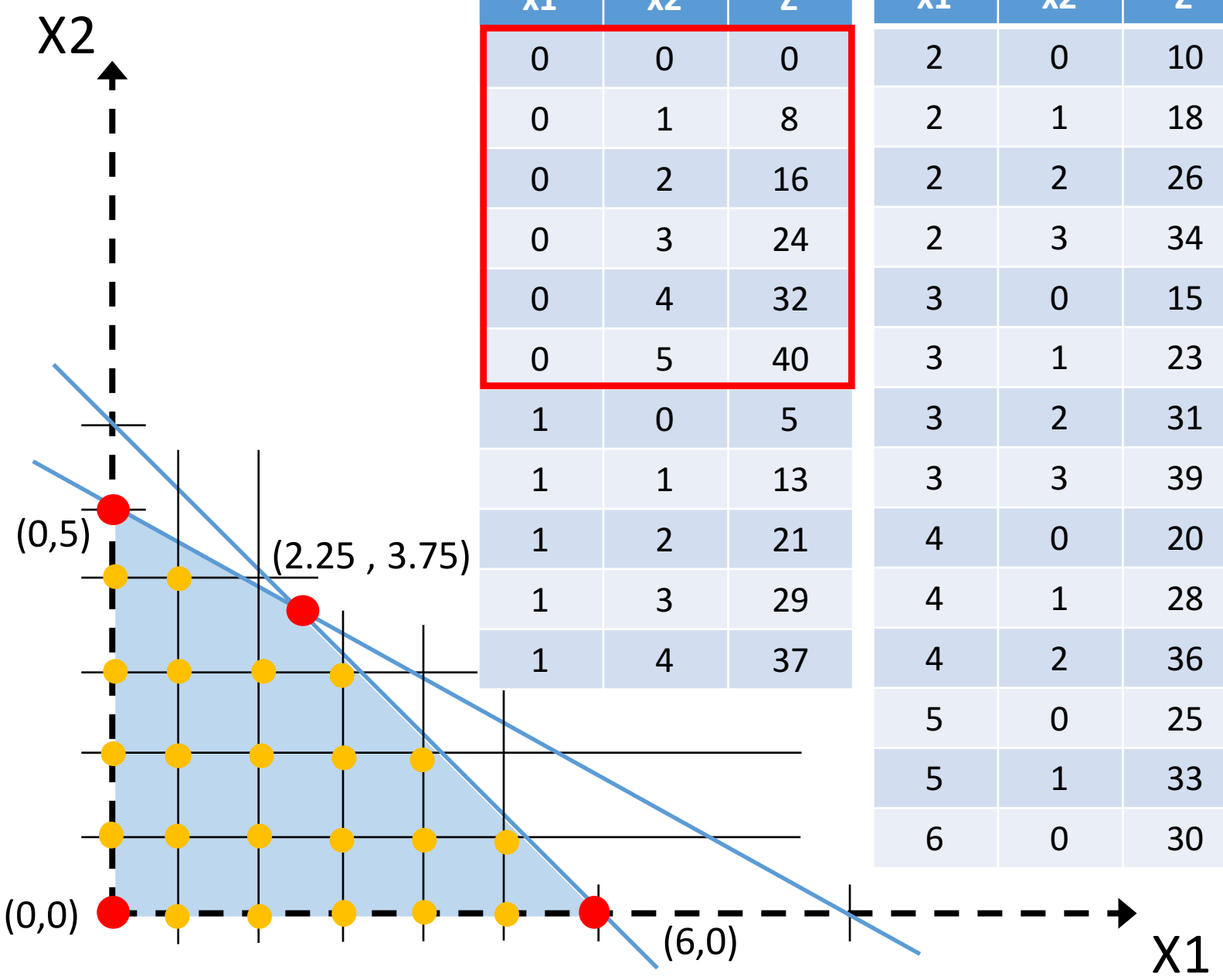
x1	x2	Z
2	0	10
2	1	18
2	2	26
2	3	34
3	0	15
3	1	23
3	2	31
3	3	39
4	0	20
4	1	28
4	2	36
5	0	25
5	1	33
6	0	30

$Z = 5x1 + 8x2$ **41.25**

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x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
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x1	x2	Z
0	0	0
0	1	8
0	2	16
0	3	24
0	4	32
0	5	40
1	0	5
1	1	13
1	2	21
1	3	29
1	4	37

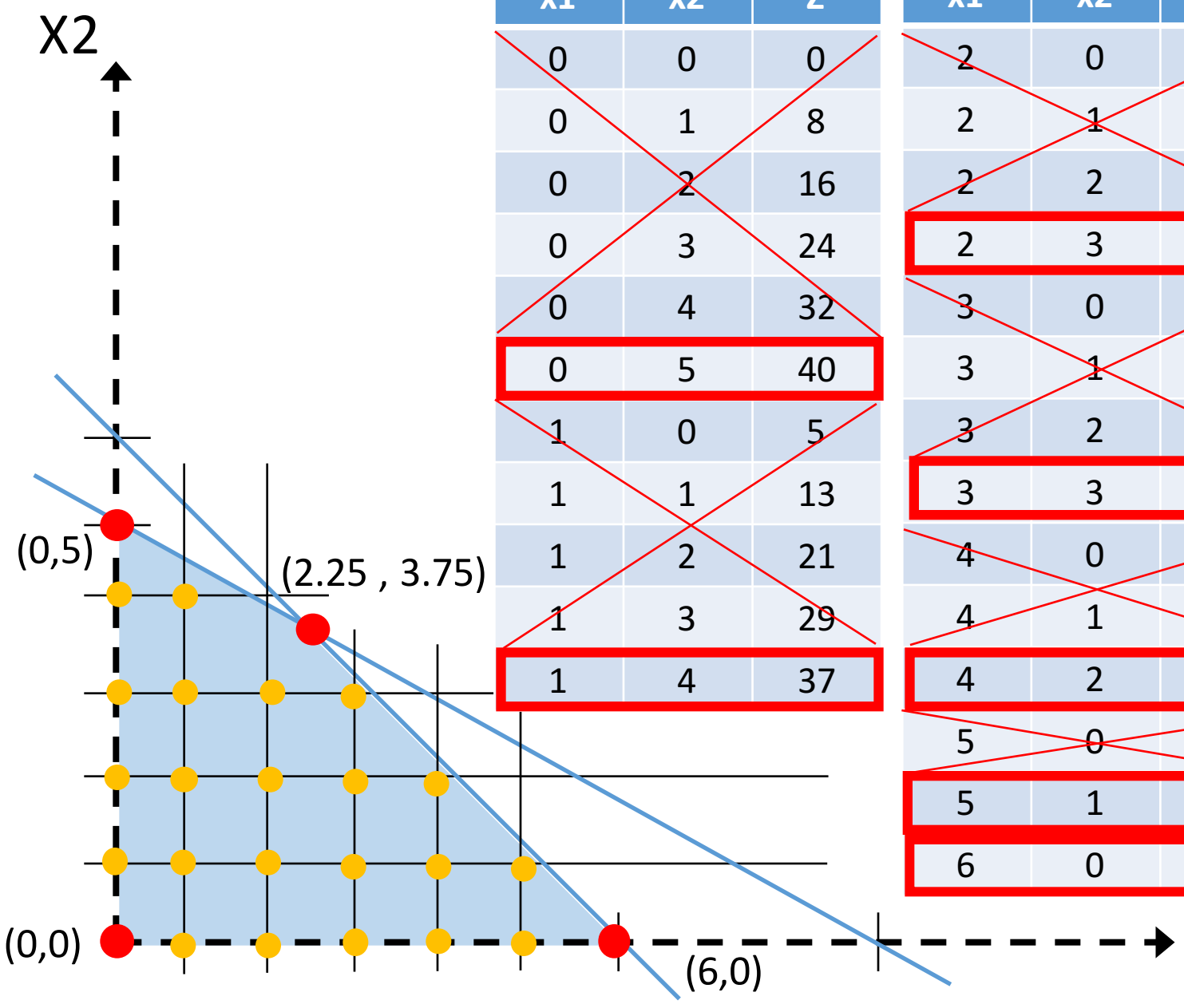
x1	x2	Z
2	0	10
2	1	18
2	2	26
2	3	34
3	0	15
3	1	23
3	2	31
3	3	39
4	0	20
4	1	28
4	2	36
5	0	25
5	1	33
6	0	30

$Z = 5x1 + 8x2$ **41.25**

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x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
6	0	30



x1	x2	Z
0	0	0
0	1	8
0	2	16
0	3	24
0	4	32
0	5	40
1	0	5
1	1	13
1	2	21
1	3	29
1	4	37

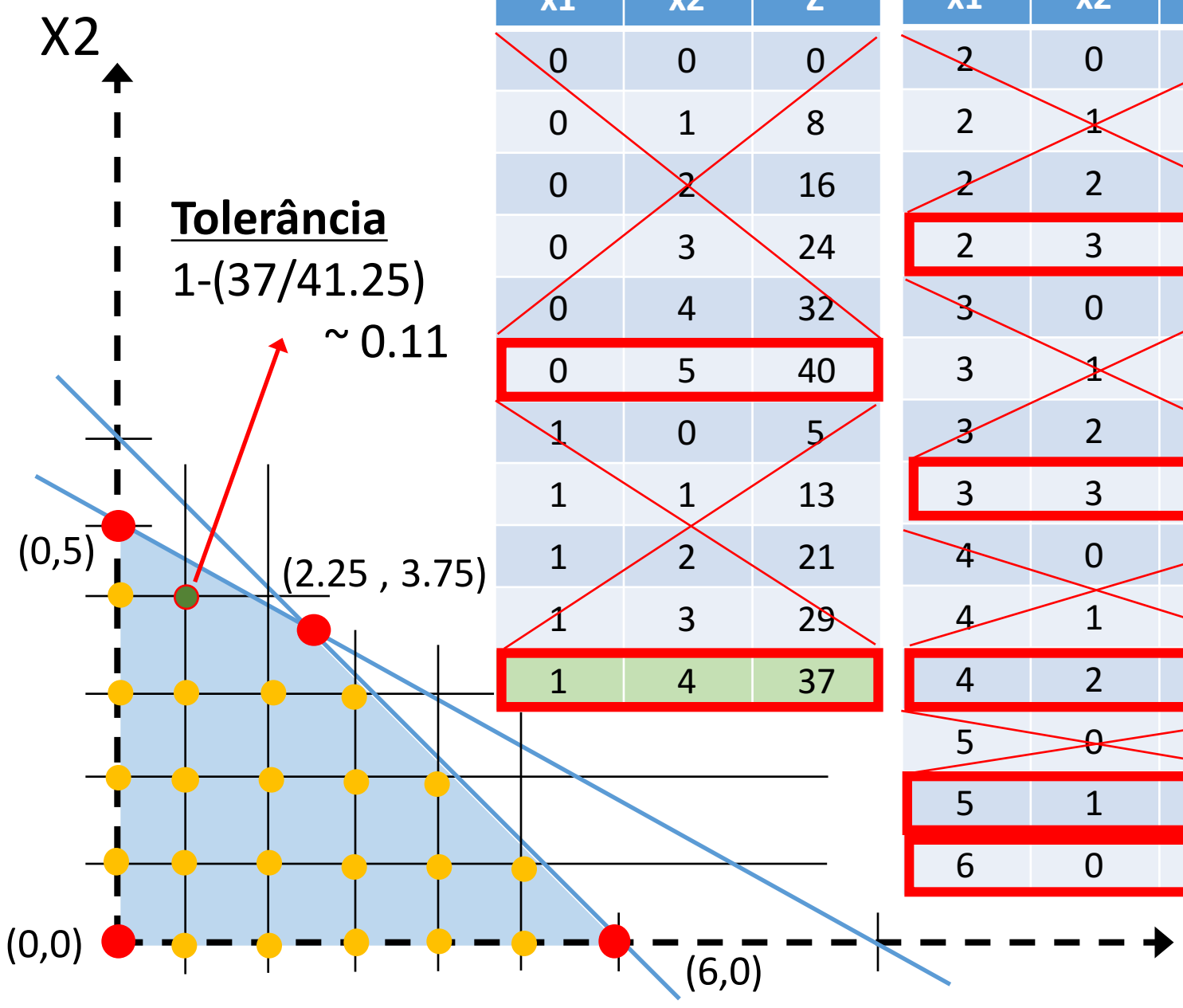
x1	x2	Z
2	0	10
2	1	18
2	2	26
2	3	34
3	0	15
3	1	23
3	2	31
3	3	39
4	0	20
4	1	28
4	2	36
5	0	25
5	1	33
6	0	30

$Z = 5x1 + 8x2$ **41.25**

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x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
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x1	x2	Z
0	0	0
0	1	8
0	2	16
0	3	24
0	4	32
0	5	40
1	0	5
1	1	13
1	2	21
1	3	29
1	4	37

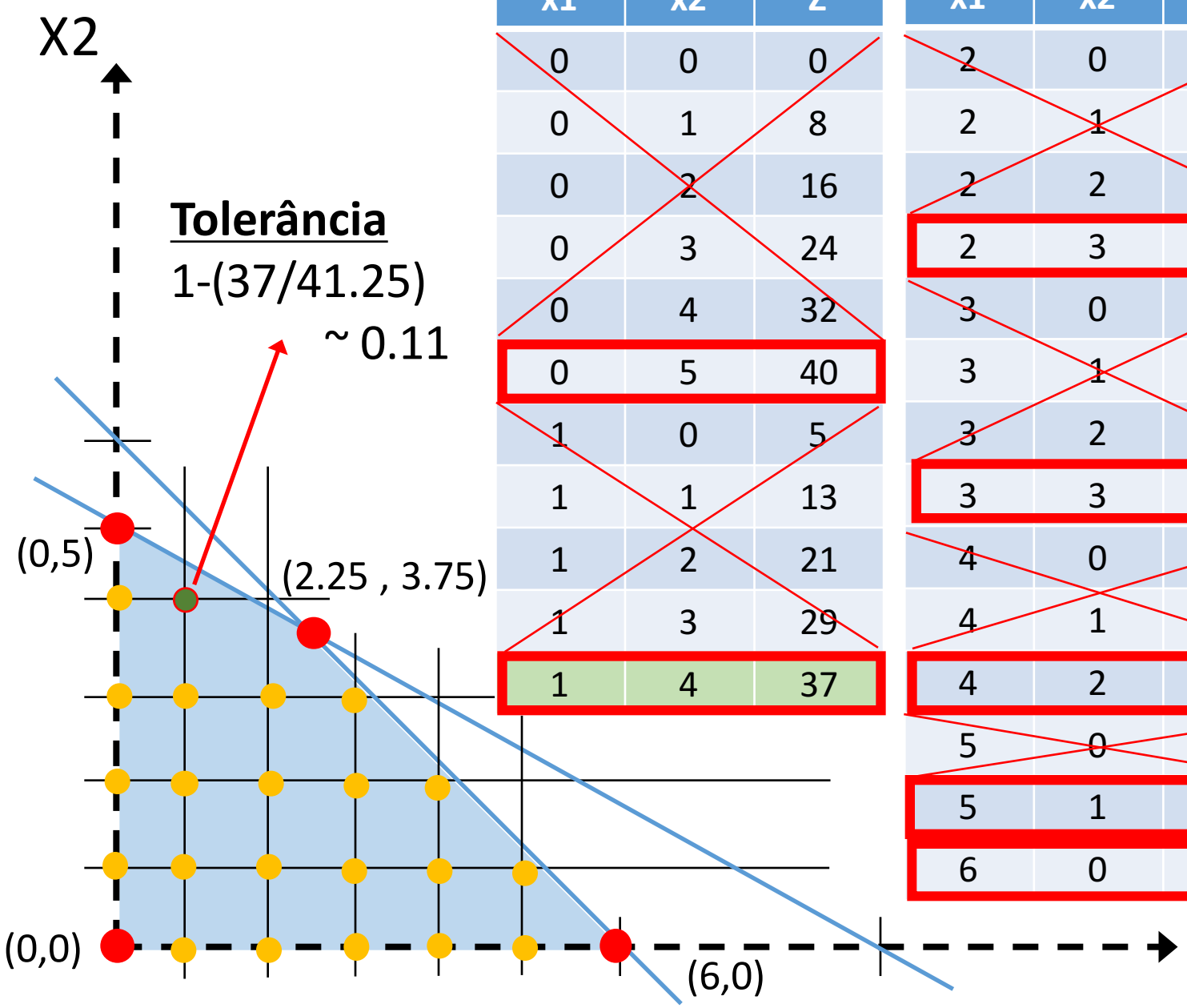
x1	x2	Z
2	0	10
2	1	18
2	2	26
2	3	34
3	0	15
3	1	23
3	2	31
3	3	39
4	0	20
4	1	28
4	2	36
5	0	25
5	1	33
6	0	30

$Z = 5x1 + 8x2$ **41.25**

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
6	0	30



x1	x2	Z
0	0	0
0	1	8
0	2	16
0	3	24
0	4	32
0	5	40
1	0	5
1	1	13
1	2	21
1	3	29
1	4	37

x1	x2	Z
2	0	10
2	1	18
2	2	26
2	3	34
3	0	15
3	1	23
3	2	31
3	3	39
4	0	20
4	1	28
4	2	36
5	0	25
5	1	33
6	0	30

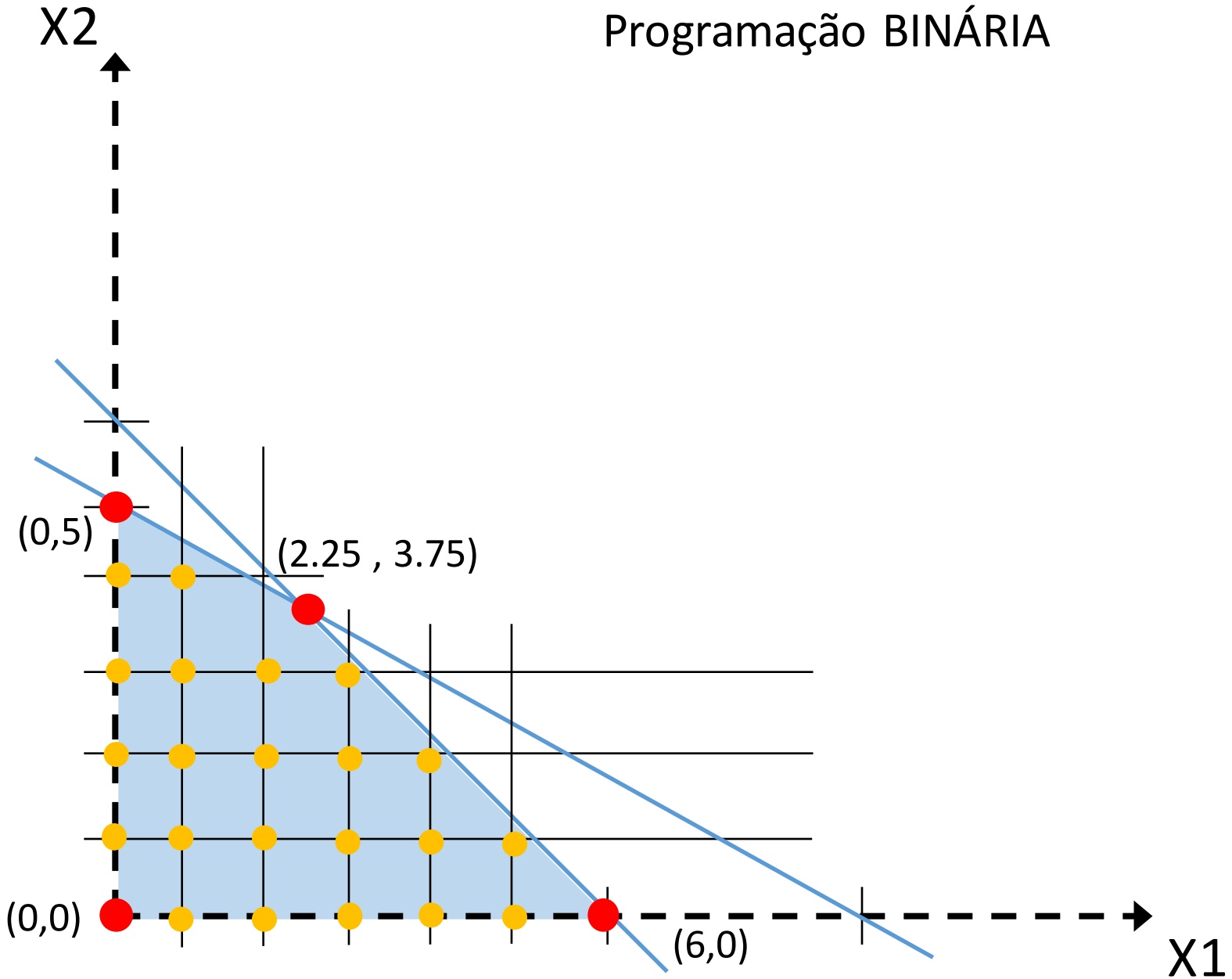
Programação BINÁRIA

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$X1 + x2 \leq 6$

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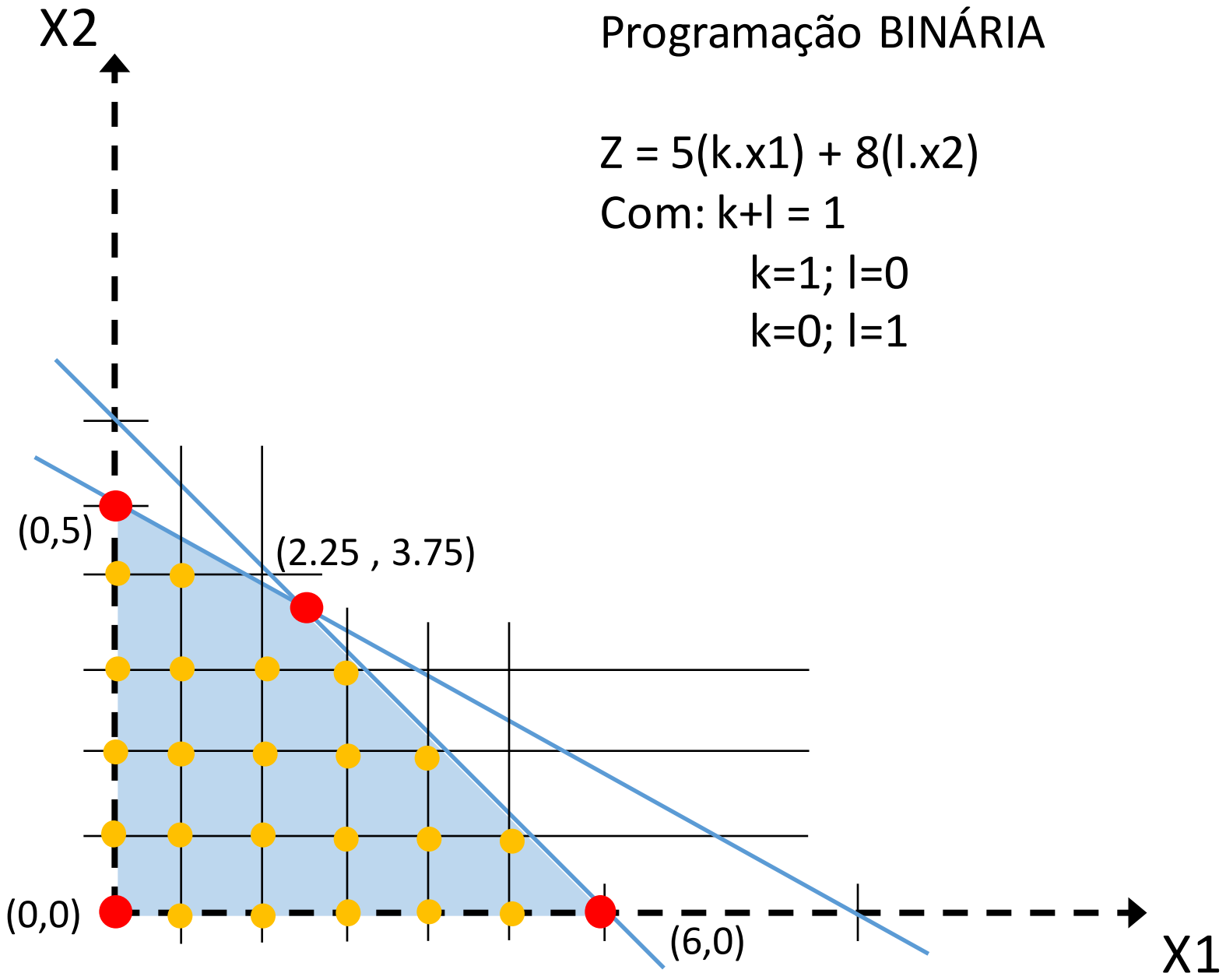


$Z = 5x1 + 8x2$ **41.25**

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x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
6	0	30



Programação BINÁRIA

$Z = 5(k.x1) + 8(l.x2)$

Com: $k+l = 1$

$k=1; l=0$

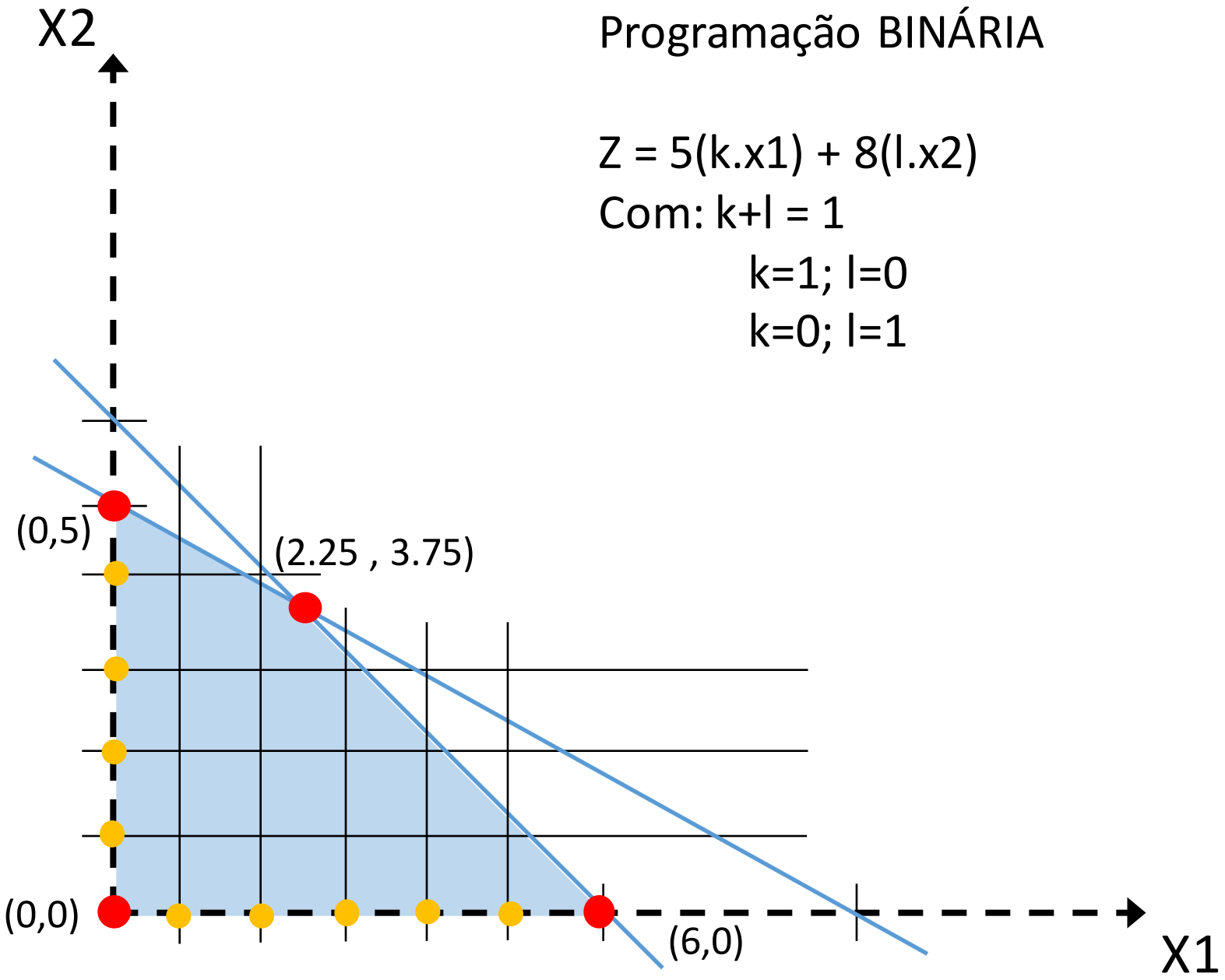
$k=0; l=1$

$Z = 5x1 + 8x2$ **41.25**

$X1 + x2 \leq 6$

$5x1 + 9x2 \leq 45$

x1	x2	Z
0	0	0
0	5	40
2.25	**3.75**	**41.25**
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Programação BINÁRIA

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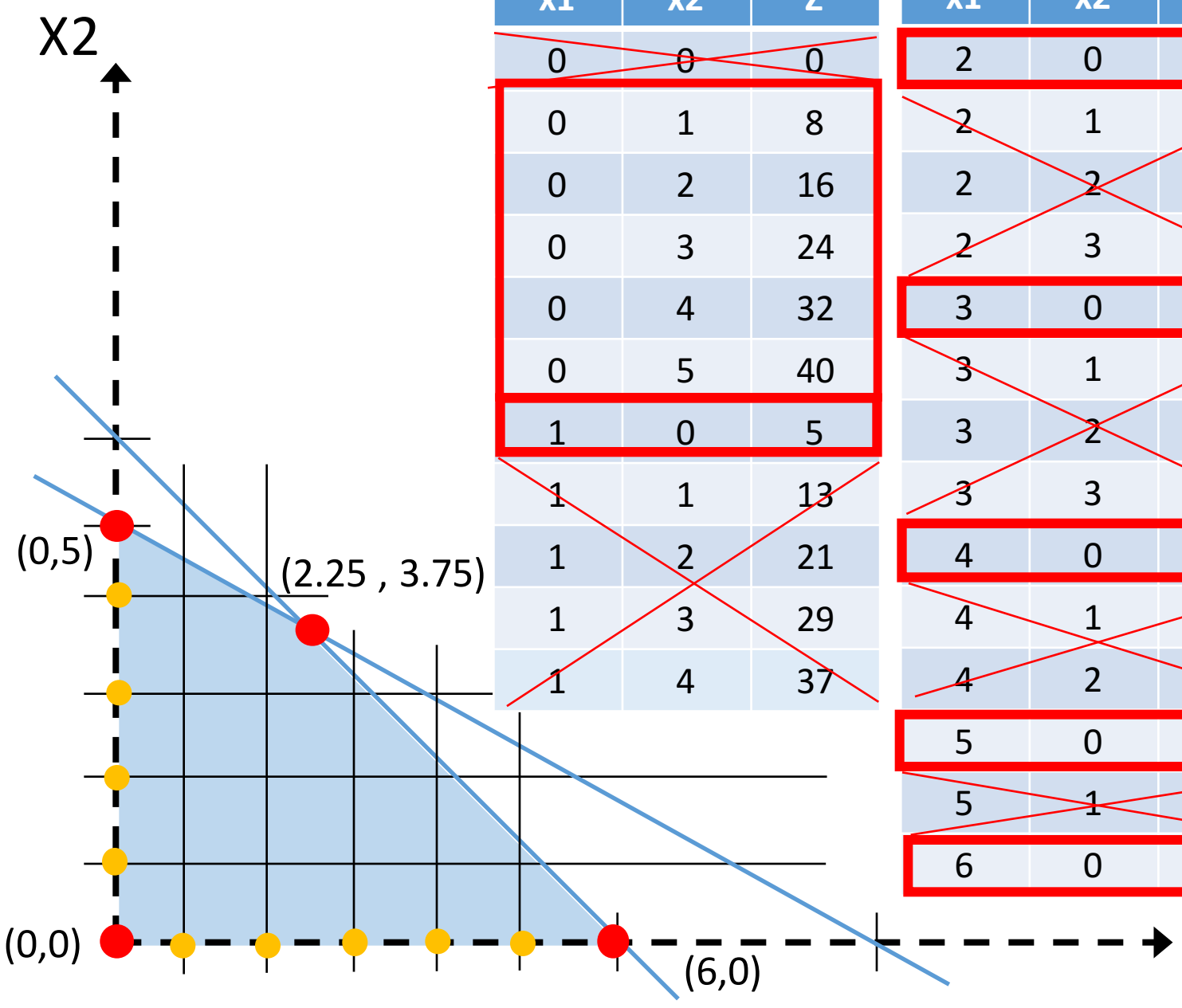
$k=0; l=1$

$Z = 5x1 + 8x2$ **41.25**

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$5x1 + 9x2 \leq 45$

x1	x2	Z
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3	3	39
4	0	20
4	1	28
4	2	36
5	0	25
5	1	33
6	0	30

MAX $Z = 3x1 + 6x2$

$X1 + x2 \geq 2$

$6x1 + 8x2 \leq 48$

$X1 \leq 6$

$X2 \leq 4$

- Qual a solução ótima?
- Arredondar a solução ótima com tolerância menor que 20%
- Obter qualquer solução inteira (dentro do espaço de soluções) com tolerância menor que 20%
- Obter a maior solução binária

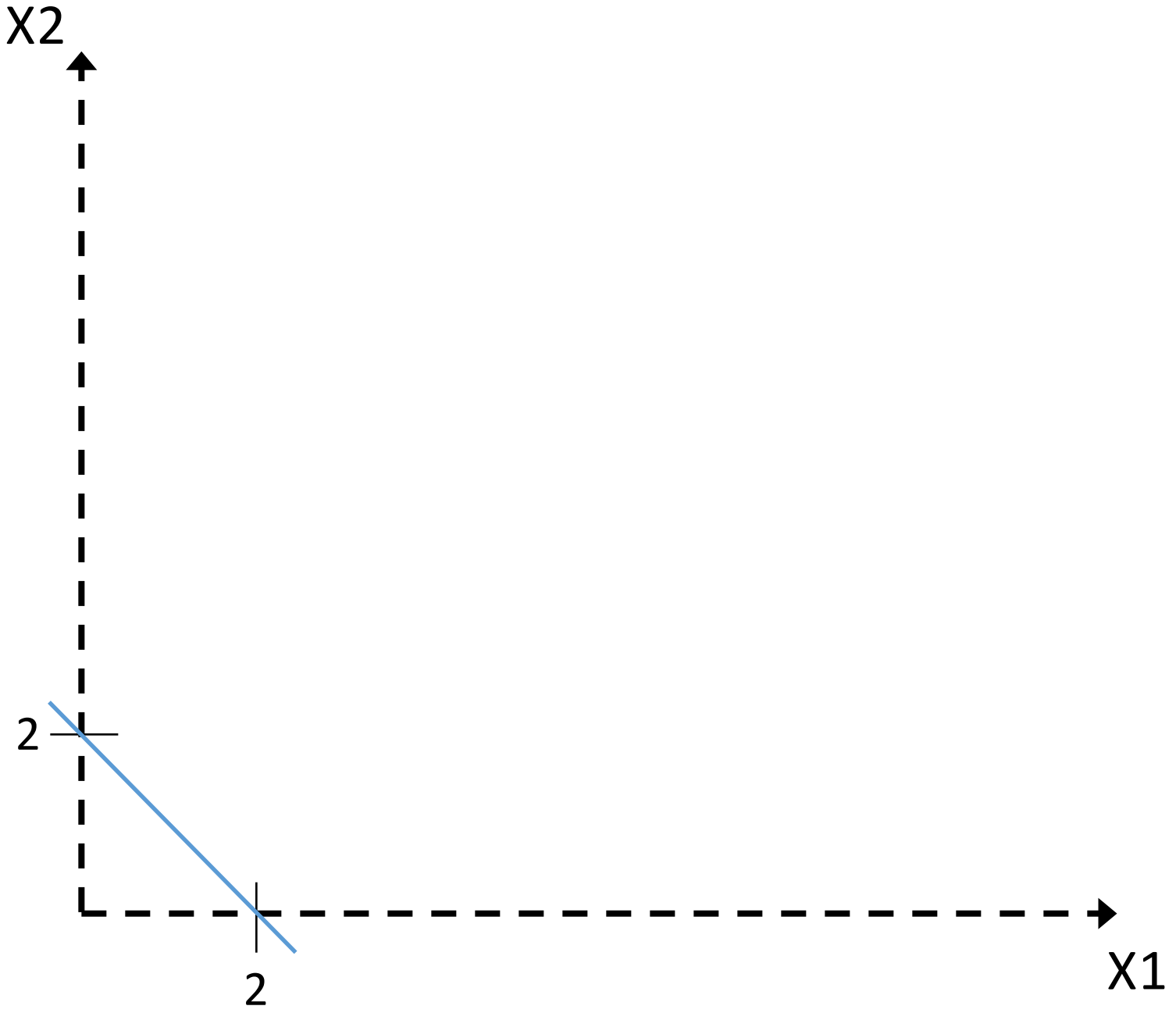
MAX Z = 3x1 + 6x2

X1 + x2 ≥ 2

6x1 + 8x2 ≤ 48

X1 ≤ 6

X2 ≤ 4



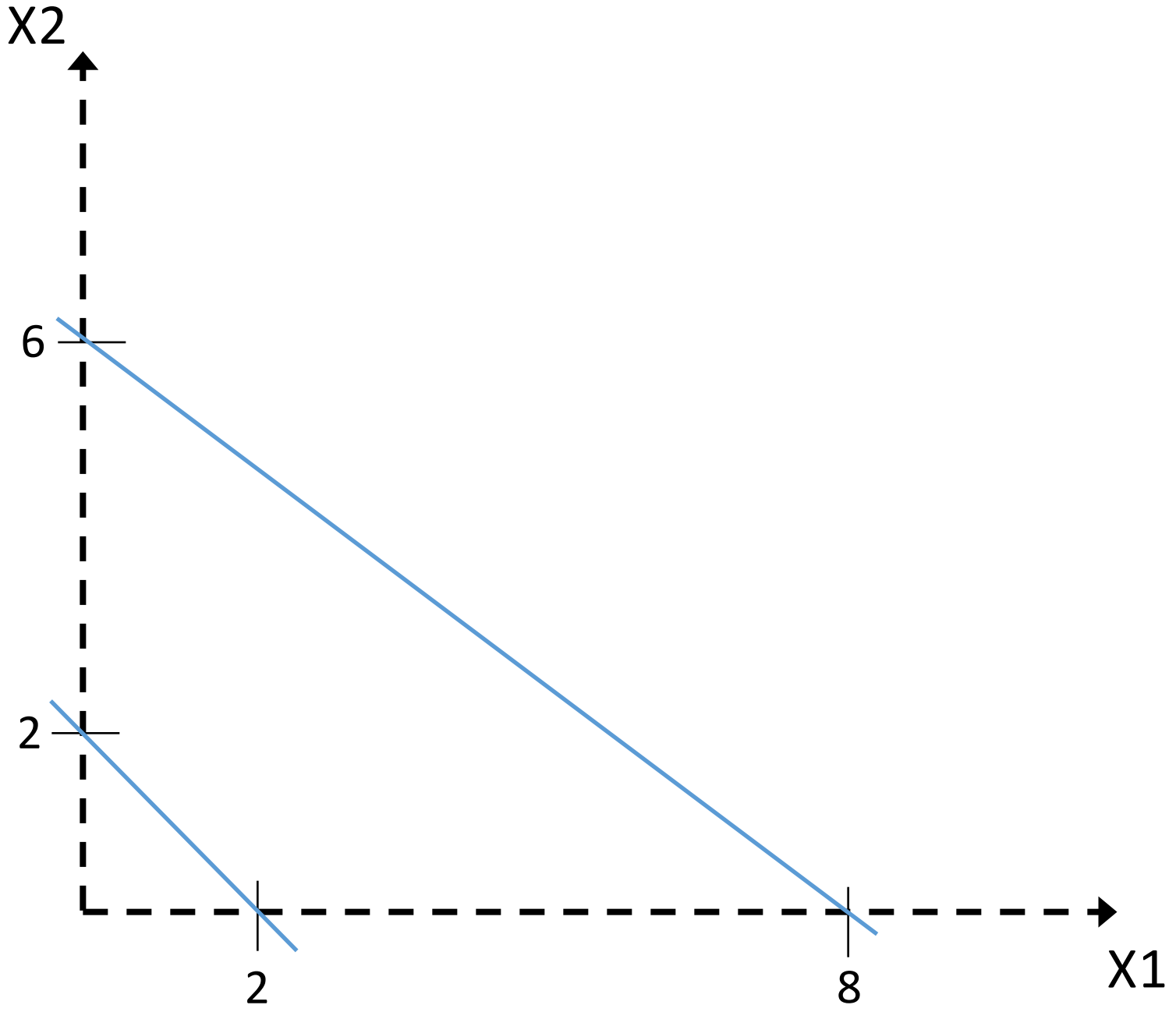
MAX Z = 3x1 + 6x2

X1 + x2 ≥ 2

6x1 + 8x2 ≤ 48

X1 ≤ 6

X2 ≤ 4



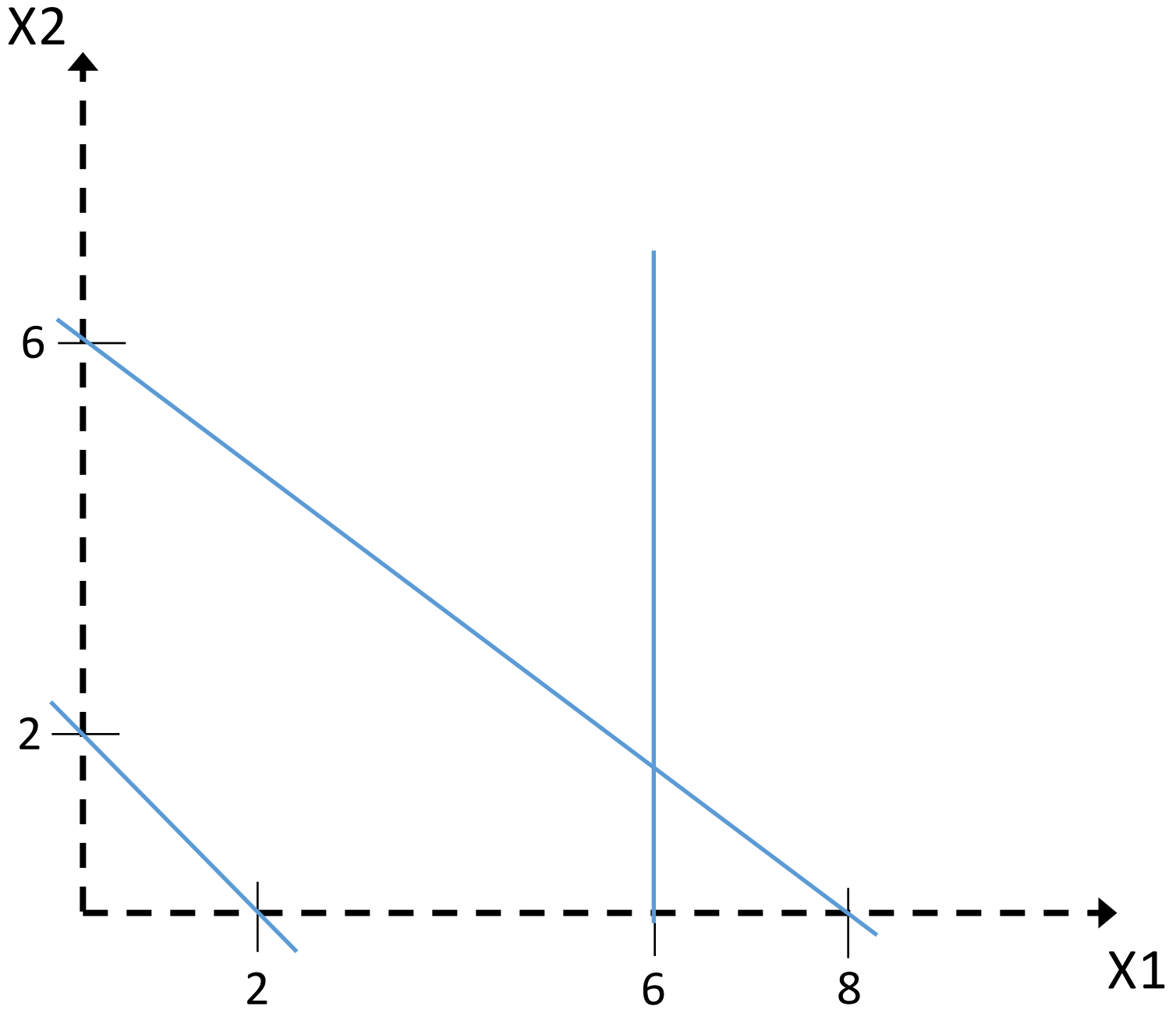
MAX Z = 3x1 + 6x2

X1 + x2 ≥ 2

6x1 + 8x2 ≤ 48

X1 ≤ 6

X2 ≤ 4



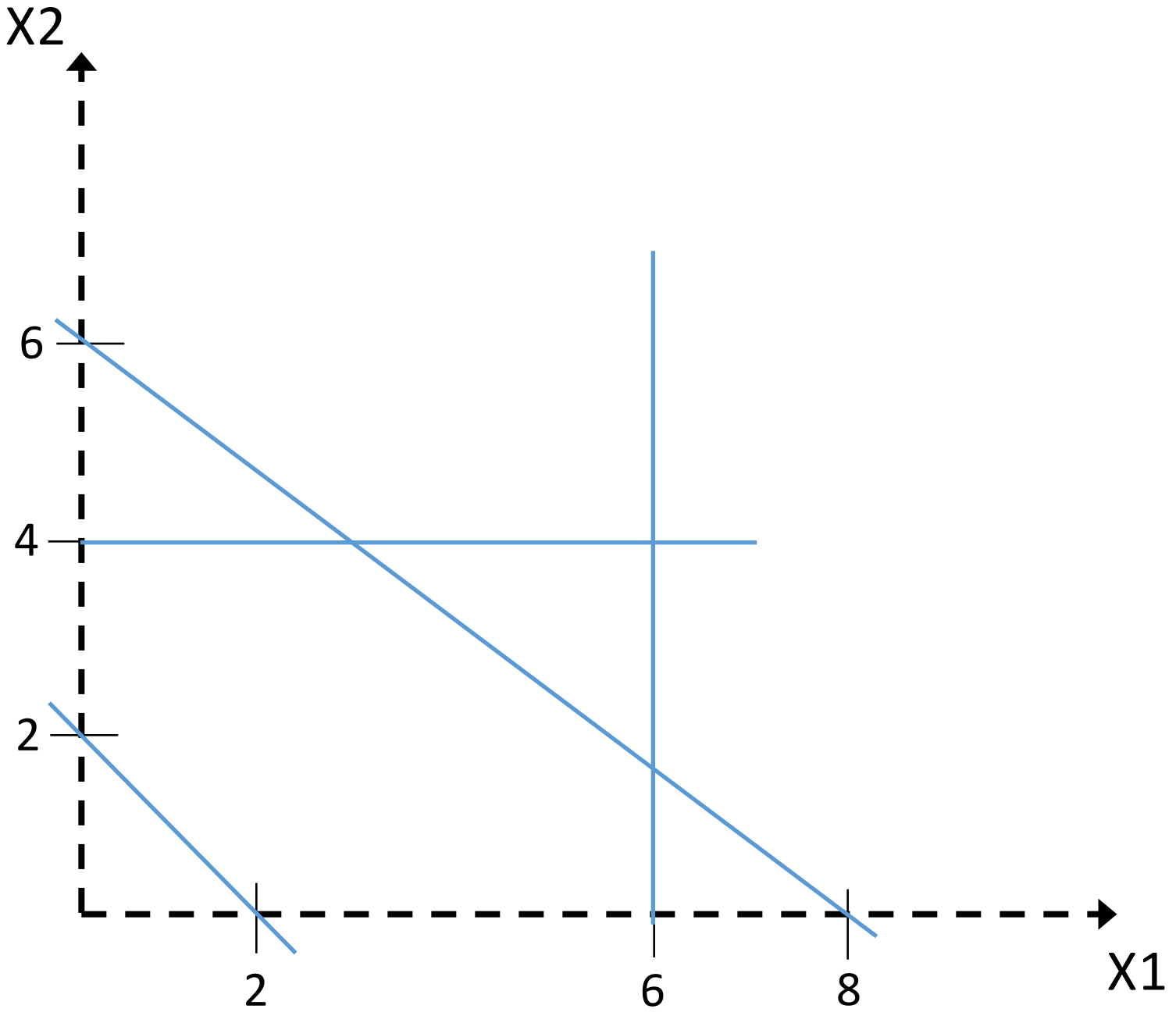
MAX Z = 3x1 + 6x2

X1 + x2 ≥ 2

6x1 + 8x2 ≤ 48

X1 ≤ 6

X2 ≤ 4



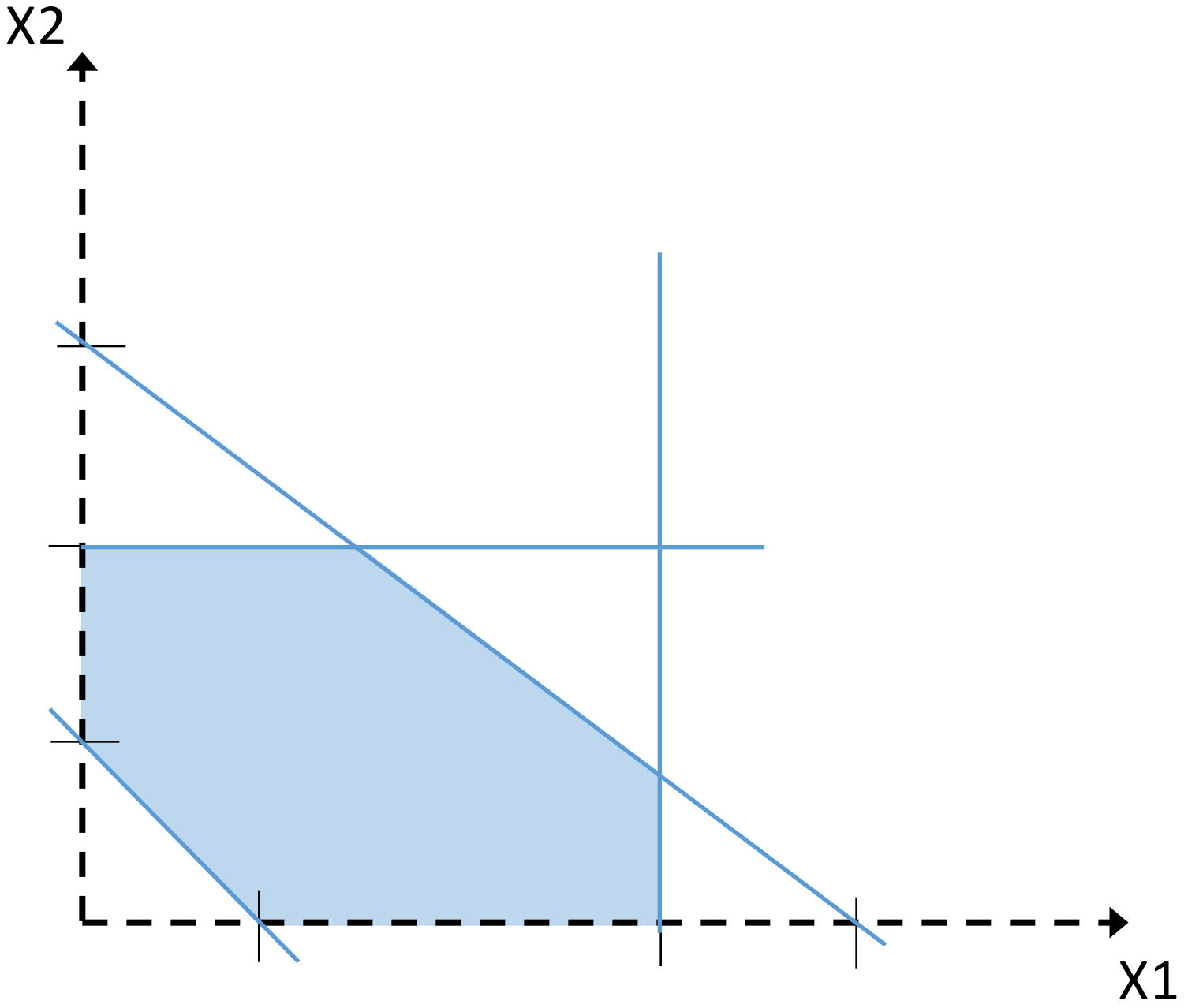
MAX Z = 3x1 + 6x2

X1 + x2 ≥ 2

6x1 + 8x2 ≤ 48

X1 ≤ 6

X2 ≤ 4



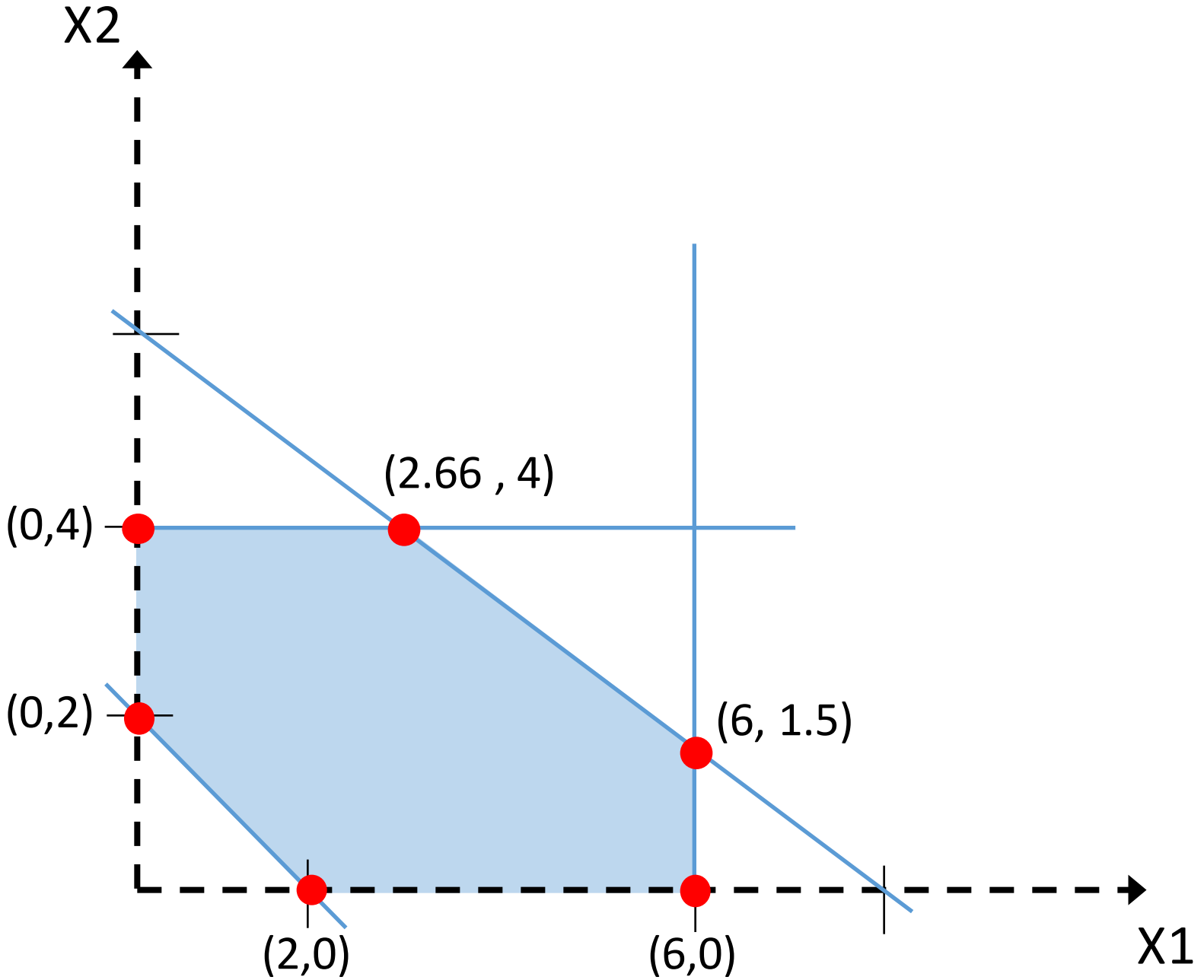
MAX Z = 3x1 + 6x2

X1 + x2 ≥ 2

6x1 + 8x2 ≤ 48

X1 ≤ 6

X2 ≤ 4



MAX Z = 3x1 + 6x2

X1 + x2 ≥ 2

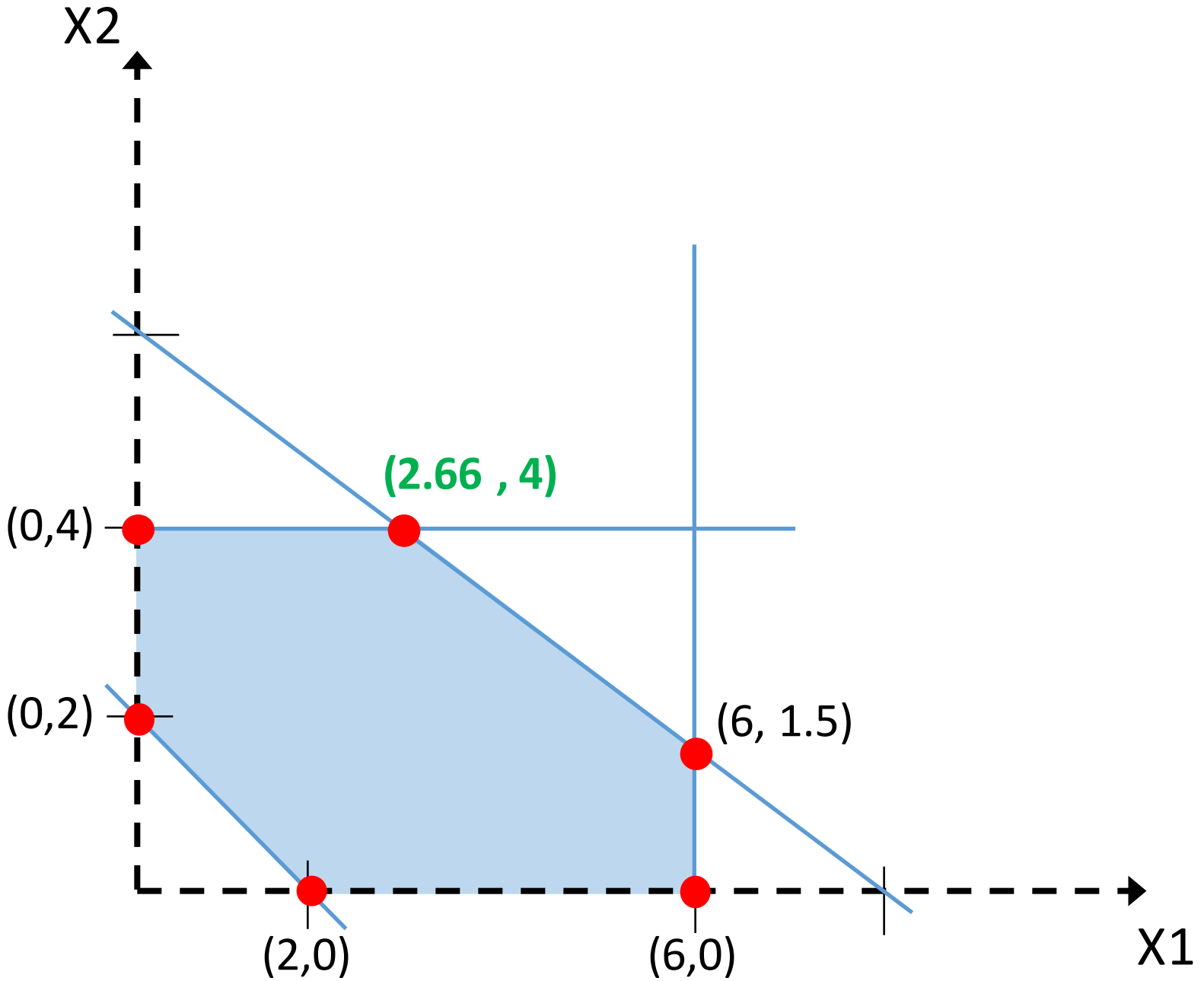
6x1 + 8x2 ≤ 48

X1 ≤ 6

X2 ≤ 4

a) Z = 3(2.66) + 6(4)

Z = 31.98



MAX Z = 3x1 + 6x2

X1 + x2 ≥ 2

6x1 + 8x2 ≤ 48

X1 ≤ 6

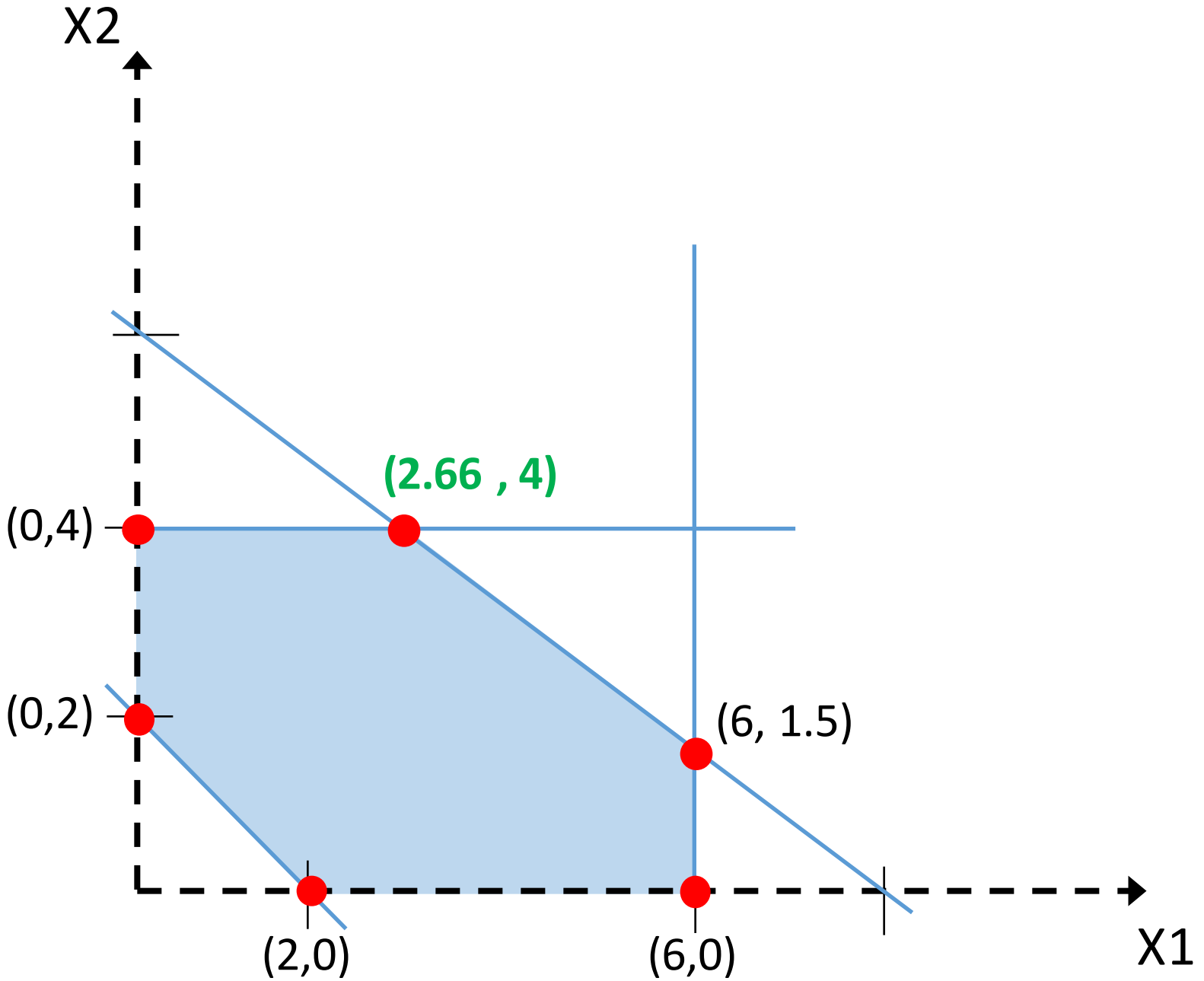
X2 ≤ 4

a) Z = 3(2.66) + 6(4)

Z = 31.98

b) Z = 3(2) + 6(4) = 30

T = 1-(30/31.98) = 6.2%



MAX Z = 3x1 + 6x2

X1 + x2 ≥ 2

6x1 + 8x2 ≤ 48

X1 ≤ 6

X2 ≤ 4

a) Z = 3(2.66) + 6(4)

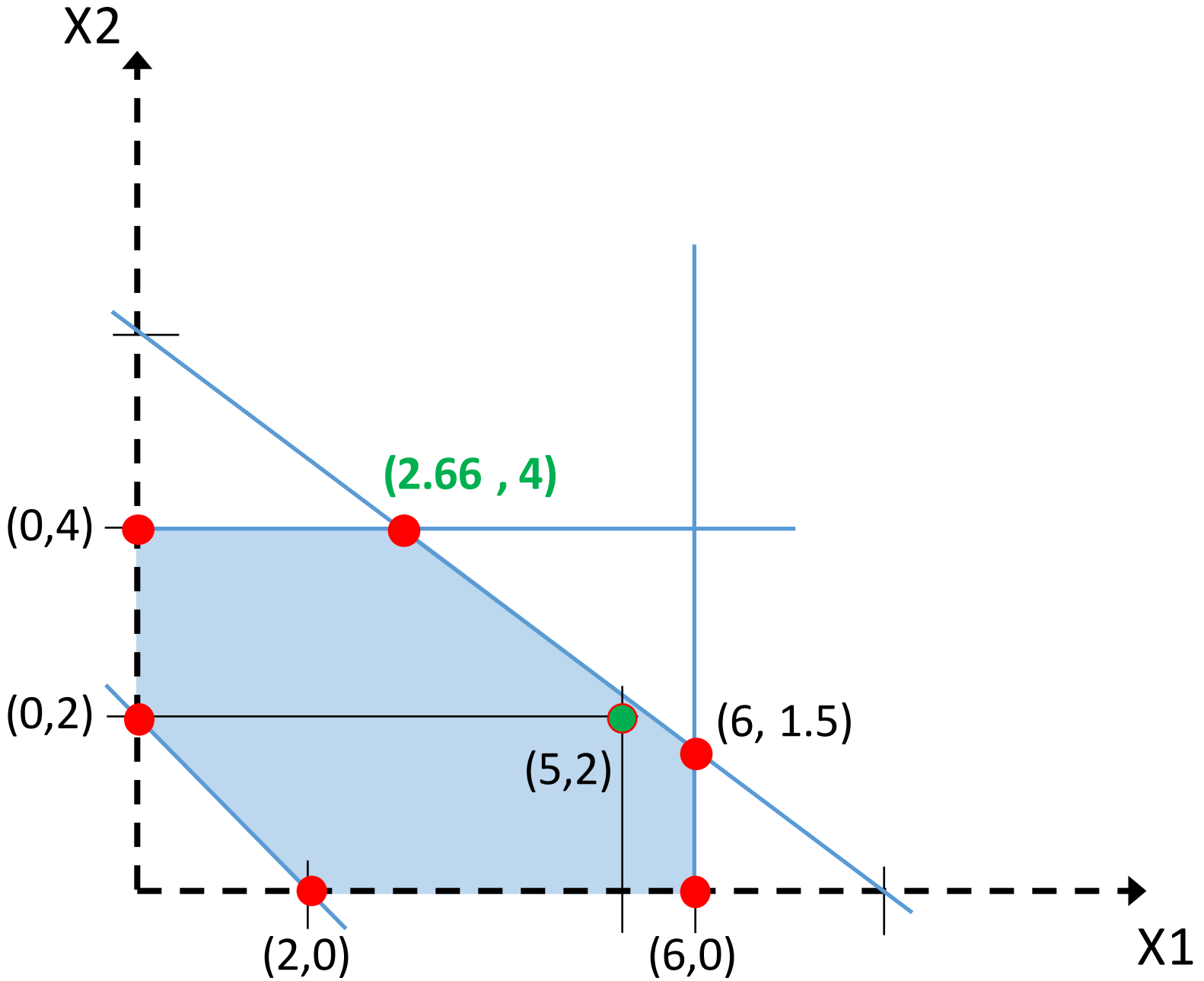
Z = 31.98

b) Z = 3(2) + 6(4) = 30

T = 1-(30/31.98) = 6.2%

c) Z = 3(5) + 6(2) = 27

T = 1-(27/31.98) = 15.5%



MAX Z = 3x1 + 6x2

X1 + x2 ≥ 2

6x1 + 8x2 ≤ 48

X1 ≤ 6

X2 ≤ 4

a) Z = 3(2.66) + 6(4)

Z = 31.98

b) Z = 3(2) + 6(4) = 30

T = 1-(30/31.98) = 6.2%

c) Z = 3(5) + 6(2) = 27

T = 1-(27/31.98) = 15.5%

d) Z = 3(0)+6(4) = 24

